popgenVCF quickstart example: chromosome 22 subset (160 samples, 8 populations)

popgenVCF

2026-08-08

Contents

1 Executive summary

This PDF file is a portable, human-readable analysis report. Figures are embedded directly in the file. The HTML version can be opened in a standard web browser, while the PDF version is optimized for compact exchange, printing, and email. The accompanying `analysis_results.rds` file is a reproducibility archive for R users; it is not required to read this report.

The analysis retained **160 samples** from **8 populations**, with **1969 QC-passing SNPs** and **357 LD-pruned SNPs**.

The global Weir-Cockerham F_{ST} estimate was **0.0915**.

2 Quality control

step	variants	removed_at_step	retained_percent
Input biallelic	21418	0	100.00
After MAF	1969	19449	9.19
After missingness	1969	0	9.19
After LD pruning	357	1612	1.67

3 Genetic diversity

population	n_samples	n_loci	polymorphic_loci	polymorphic_fraction	observed_heterozygosity	expected_heterozygosity	inbreeding_coefficient	mean_minor_allele_frequency	mean_locus_call_rate	hwe_tested_loci	hwe_significant_loci	hwe_significant_loci_fdr	private_allele_loci
CHB	20	1969	1653	0.8395	0.2683	0.2855	0.0603	0.2133	1	1653	149	55	0
GBR	20	1969	1736	0.8817	0.2764	0.2827	0.0226	0.2004	1	1736	99	31	0
ITU	20	1969	1829	0.9289	0.2736	0.2919	0.0626	0.2059	1	1829	100	43	0
LWK	20	1969	1622	0.8238	0.2638	0.2727	0.0326	0.1963	1	1622	66	14	0
PEL	20	1969	1800	0.9142	0.2330	0.2579	0.0968	0.1851	1	1800	147	45	0
PUR	20	1969	1844	0.9365	0.2748	0.2730	-0.0067	0.1871	1	1844	81	21	0
STU	20	1969	1788	0.9081	0.2561	0.2863	0.1056	0.2013	1	1788	131	33	0
YRI	20	1969	1636	0.8309	0.2731	0.2705	-0.0096	0.1944	1	1636	92	18	0

4 Principal component analysis

sample	vcf_sample	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8	PC9	PC10	population
NA18536	NA18536	-0.0575	0.0756	-0.1958	-0.0176	0.0470	0.0646	-0.0320	-0.0509	-0.0829	0.0169	CHB
NA18538	NA18538	-0.0665	0.1310	-0.0958	-0.0006	0.0158	-0.0224	0.1012	0.0074	-0.2047	-0.0051	CHB
NA18541	NA18541	-0.0654	0.0991	-0.0443	-0.0362	0.0684	0.0917	0.0875	-0.0995	-0.0353	0.0002	CHB
NA18552	NA18552	-0.0714	0.1650	-0.1811	-0.1448	0.0223	0.0384	0.0671	0.0537	0.0328	-0.0593	CHB
NA18553	NA18553	-0.0616	-0.0660	-0.1590	0.0237	-0.2193	-0.0004	-0.1910	-0.0064	-0.1380	0.0197	CHB
NA18555	NA18555	-0.0312	0.0930	-0.1309	-0.1100	0.0308	0.0717	-0.0562	0.0124	0.0901	-0.1218	CHB
NA18557	NA18557	-0.0893	0.2057	-0.1264	-0.0737	0.0509	0.0121	0.0892	0.0377	-0.0913	0.0984	CHB
NA18562	NA18562	-0.0381	0.0209	-0.0813	0.0459	0.0625	-0.0312	0.1107	-0.1086	0.0245	-0.1299	CHB
NA18564	NA18564	-0.0305	0.0841	-0.1174	-0.1036	-0.0883	-0.0264	-0.1778	0.0857	0.2782	-0.1910	CHB
NA18573	NA18573	-0.0685	0.0633	0.0378	0.0394	-0.0765	0.0588	-0.0318	-0.1708	0.0201	-0.0102	CHB
NA18574	NA18574	-0.0873	0.1260	-0.1222	-0.0647	0.0656	0.0098	0.0426	-0.0156	-0.0750	-0.0113	CHB
NA18606	NA18606	-0.0584	0.0752	-0.1100	0.0288	0.0297	-0.0415	0.1097	0.1042	-0.0327	-0.0739	CHB
NA18611	NA18611	-0.0717	0.1273	-0.0419	-0.0468	-0.0176	0.1518	-0.0208	0.0271	-0.1508	-0.0482	CHB
NA18612	NA18612	-0.0504	-0.0424	-0.2118	0.1398	-0.1597	-0.0342	-0.1189	-0.0957	-0.0276	0.1465	CHB
NA18613	NA18613	-0.0583	0.1108	-0.0705	-0.0469	0.0092	0.0932	0.0875	0.0691	-0.1390	0.1136	CHB
NA18616	NA18616	-0.0173	0.0510	-0.1403	-0.1915	0.0842	0.1383	-0.0259	-0.1606	0.0728	-0.0742	CHB
NA18617	NA18617	-0.0307	0.1223	-0.1382	-0.1085	-0.0277	0.0449	-0.0945	0.0113	0.0748	-0.1828	CHB
NA18638	NA18638	-0.0722	0.1224	-0.0679	-0.0460	-0.0106	0.1791	-0.0069	-0.1120	-0.1554	0.0068	CHB
NA18643	NA18643	-0.0437	0.0467	-0.1922	0.0077	0.0101	-0.0009	0.0329	-0.0462	0.0704	-0.0114	CHB
NA18647	NA18647	-0.0558	0.1011	-0.1989	-0.0603	0.1633	0.1660	0.0094	-0.0507	0.0072	0.0560	CHB

5 PCA SNP loadings (top 20 by |loading|; see 31_PCA_loadings.tsv for the complete top-N list)

axis	snp_id	chromosome	position	contribution	magnitude
PC7	10937	22	20437004	0.2848	0.2848
PC7	10502	22	20371297	0.2541	0.2541
PC7	11374	22	20502771	0.2524	0.2524
PC8	13593	22	20753041	0.2191	0.2191
PC6	10937	22	20437004	-0.2191	0.2191
PC8	13990	22	20765538	0.2178	0.2178
PC8	14246	22	20773711	0.2034	0.2034
PC5	4933	22	20159049	-0.2010	0.2010
PC3	15906	22	20832523	0.1980	0.1980
PC8	13078	22	20733667	0.1952	0.1952
PC7	10415	22	20360856	0.1932	0.1932
PC8	14080	22	20768038	-0.1925	0.1925
PC9	3724	22	20121080	-0.1900	0.1900
PC6	10502	22	20371297	-0.1887	0.1887
PC10	11570	22	20630754	0.1883	0.1883
PC9	1516	22	20047587	-0.1869	0.1869
PC3	14848	22	20794119	0.1865	0.1865
PC9	2006	22	20065314	-0.1859	0.1859
PC8	14377	22	20777533	-0.1806	0.1806
PC10	11593	22	20639324	0.1763	0.1763

6 Population differentiation

population_1	population_2	n_1	n_2	fst
CHB	GBR	20	20	0.1210
CHB	ITU	20	20	0.0631
CHB	LWK	20	20	0.1631
CHB	PEL	20	20	0.0794
CHB	PUR	20	20	0.1101
CHB	STU	20	20	0.0865
CHB	YRI	20	20	0.1631
GBR	ITU	20	20	0.0232
GBR	LWK	20	20	0.1198
GBR	PEL	20	20	0.0874
GBR	PUR	20	20	0.0267
GBR	STU	20	20	0.0350
GBR	YRI	20	20	0.1344
ITU	LWK	20	20	0.1057
ITU	PEL	20	20	0.0400
ITU	PUR	20	20	0.0269
ITU	STU	20	20	0.0052
ITU	YRI	20	20	0.1189
LWK	PEL	20	20	0.1737
LWK	PUR	20	20	0.1176
LWK	STU	20	20	0.1112
LWK	YRI	20	20	0.0052
PEL	PUR	20	20	0.0621
PEL	STU	20	20	0.0609
PEL	YRI	20	20	0.1799
PUR	STU	20	20	0.0339
PUR	YRI	20	20	0.1226
STU	YRI	20	20	0.1187

7 Genome scan FST outliers (top 20 windows by FST; see 39_genome_scan_fst.tsv for the complete list)

chromosome	window_start	window_end	n_snps	global_fst
22	20800428	20850427	104	0.1718
22	20750428	20800427	213	0.1319
22	20850428	20900427	113	0.1118
22	20250428	20300427	102	0.1097
22	20900428	20950427	138	0.1076
22	20150428	20200427	196	0.1007
22	20200428	20250427	111	0.0952
22	20000428	20050427	160	0.0875
22	20700428	20750427	135	0.0845
22	20400428	20450427	24	0.0838
22	20100428	20150427	115	0.0694
22	20950428	21000427	193	0.0693
22	20350428	20400427	64	0.0666
22	20450428	20500427	48	0.0639
22	20050428	20100427	131	0.0512
22	20600428	20650427	38	0.0510
22	20500428	20550427	9	0.0403
22	20300428	20350427	44	0.0351
22	20650428	20700427	31	0.0194

8 Close relatives (up to 20 shown; see 36_close_relatives.tsv for the complete list)

sample_1	sample_2	population_1	population_2	IBS0	kinship	relationship_degree
HG03873	HG03998	ITU	STU	0.0000	0.4525	duplicate/MZ twin
NA19331	NA19334	LWK	LWK	0.0028	0.4459	duplicate/MZ twin
HG01938	HG02278	PEL	PEL	0.0056	0.2781	1st-degree
HG00742	HG01054	PUR	PUR	0.0140	0.2468	1st-degree
NA19172	NA19247	YRI	YRI	0.0168	0.2448	1st-degree
NA19360	NA18516	LWK	YRI	0.0112	0.2313	1st-degree
HG01067	HG01182	PUR	PUR	0.0140	0.2225	1st-degree
HG01067	HG01086	PUR	PUR	0.0196	0.2125	1st-degree
HG00099	HG00141	GBR	GBR	0.0112	0.2110	1st-degree
HG01067	HG01305	PUR	PUR	0.0168	0.2100	1st-degree
HG00122	HG01308	GBR	PUR	0.0140	0.1971	1st-degree
NA18516	NA19222	YRI	YRI	0.0140	0.1959	1st-degree
HG01396	HG03740	PUR	STU	0.0084	0.1944	1st-degree
NA18516	NA19210	YRI	YRI	0.0196	0.1910	1st-degree
HG00253	HG01067	GBR	PUR	0.0168	0.1900	1st-degree
HG01938	HG02345	PEL	PEL	0.0084	0.1875	1st-degree
NA18516	NA19247	YRI	YRI	0.0196	0.1856	1st-degree
HG00250	HG00253	GBR	GBR	0.0168	0.1852	1st-degree
HG00141	HG00240	GBR	GBR	0.0168	0.1825	1st-degree
HG00253	HG01396	GBR	PUR	0.0168	0.1818	1st-degree

9 Runs of homozygosity (highest FROH samples, up to 20 shown; see 38_ROH_sample_summary.tsv for the complete list)

sample	population	n_runs	total_length_bp	mean_length_bp	longest_run_bp	froh
HG02146	PEL	4	788356	197089.00	574060	0.7884
HG01942	PEL	4	595694	148923.50	235561	0.5957
HG01933	PEL	4	537370	134342.50	311897	0.5374
HG01578	PEL	6	528877	88146.17	194872	0.5289
HG03989	STU	2	469894	234947.00	265232	0.4699
HG00151	GBR	4	438830	109707.50	207650	0.4388
HG01950	PEL	4	438096	109524.00	217282	0.4381
NA18611	CHB	3	437054	145684.67	249970	0.4371
HG00637	PUR	4	407361	101840.25	155344	0.4074
HG00123	GBR	3	406869	135623.00	221114	0.4069
NA18612	CHB	3	400265	133421.67	226775	0.4003
NA18616	CHB	5	391340	78268.00	268046	0.3914
HG01054	PUR	3	379034	126344.67	228087	0.3790
HG01086	PUR	3	364023	121341.00	162866	0.3640
HG03836	STU	2	359185	179592.50	284841	0.3592
HG03990	STU	3	349620	116540.00	175138	0.3496
NA18647	CHB	3	339679	113226.33	182839	0.3397
HG01917	PEL	3	337023	112341.00	261589	0.3370
HG01927	PEL	2	330751	165375.50	191744	0.3308
NA18564	CHB	3	323282	107760.67	181407	0.3233

10 Discriminant analysis of principal components

K	n_pca	n_da	BIC	assignment_accuracy	mean_success	calinski_harabasz	davies_bouldin	replicate_max_rmse
2	40	1	NA	0.2500	0.9958	15.6223	2.7833	0.0971
3	10	2	NA	0.2938	0.9966	13.3767	3.0860	0.0000
4	10	3	NA	0.3562	0.9648	11.6552	2.8675	0.3841
5	30	4	NA	0.4062	0.9361	10.2898	2.6773	0.3512
6	10	5	NA	0.3938	0.9306	9.5564	2.5154	0.2944
7	10	6	NA	0.4812	0.9407	9.0366	2.4878	0.0727
8	10	7	NA	0.4625	0.9058	8.3942	2.4244	0.1614
9	10	8	NA	0.4688	0.9239	7.8689	2.6212	0.1830
10	10	9	NA	0.3812	0.8975	7.3769	2.6614	0.2373

11 DAPC SNP loadings (top 20 by contribution across all K; see 22f_DAPC_loadings_K.tsv for the complete per-K top-N lists)

K	axis	snp_id	chromosome	position	contribution
3	LD2	10525	22	20376408	0.0202
3	LD2	10527	22	20376436	0.0199
3	LD2	10765	22	20396584	0.0184
4	LD2	10525	22	20376408	0.0184
3	LD2	10675	22	20388698	0.0184
3	LD2	10757	22	20394994	0.0182
3	LD2	10807	22	20401090	0.0182
4	LD2	10527	22	20376436	0.0181
4	LD2	10807	22	20401090	0.0181
3	LD2	10806	22	20401066	0.0180
3	LD2	10758	22	20395128	0.0180
4	LD2	10765	22	20396584	0.0179
4	LD2	10758	22	20395128	0.0179
4	LD2	10823	22	20402404	0.0179
3	LD2	10937	22	20437004	0.0178
4	LD2	10806	22	20401066	0.0177
3	LD2	10823	22	20402404	0.0177
3	LD2	10561	22	20378905	0.0175
4	LD2	10757	22	20394994	0.0174
3	LD2	10812	22	20401719	0.0174

12 Analysis of molecular variance

component	Sigma	%
Variations Between population	27.1959	9.0406
Variations Between samples Within population	12.8319	4.2657
Variations Within samples	260.7906	86.6937
Total variations	300.8184	100.0000

statistic	Phi
Phi-samples-total	0.1331
Phi-samples-population	0.0469
Phi-population-total	0.0904

13 Mantel test and isolation by distance

mantel_r	mantel_p	slope	r_squared
0.3129	0.001	0.0028	0.1366

14 Chromosome-specific results

chromosome	qc_snps	ld_snps	global_fst	pc1_percent
22	1969	357	0.0915	50.9697

15 Figure gallery

The gallery contains one version of every analysis figure. The HTML report prefers browser-friendly SVG or PNG files. The PDF report prefers the compact vector PDF source for each figure. Original files remain available in the sibling `figures/` directory.

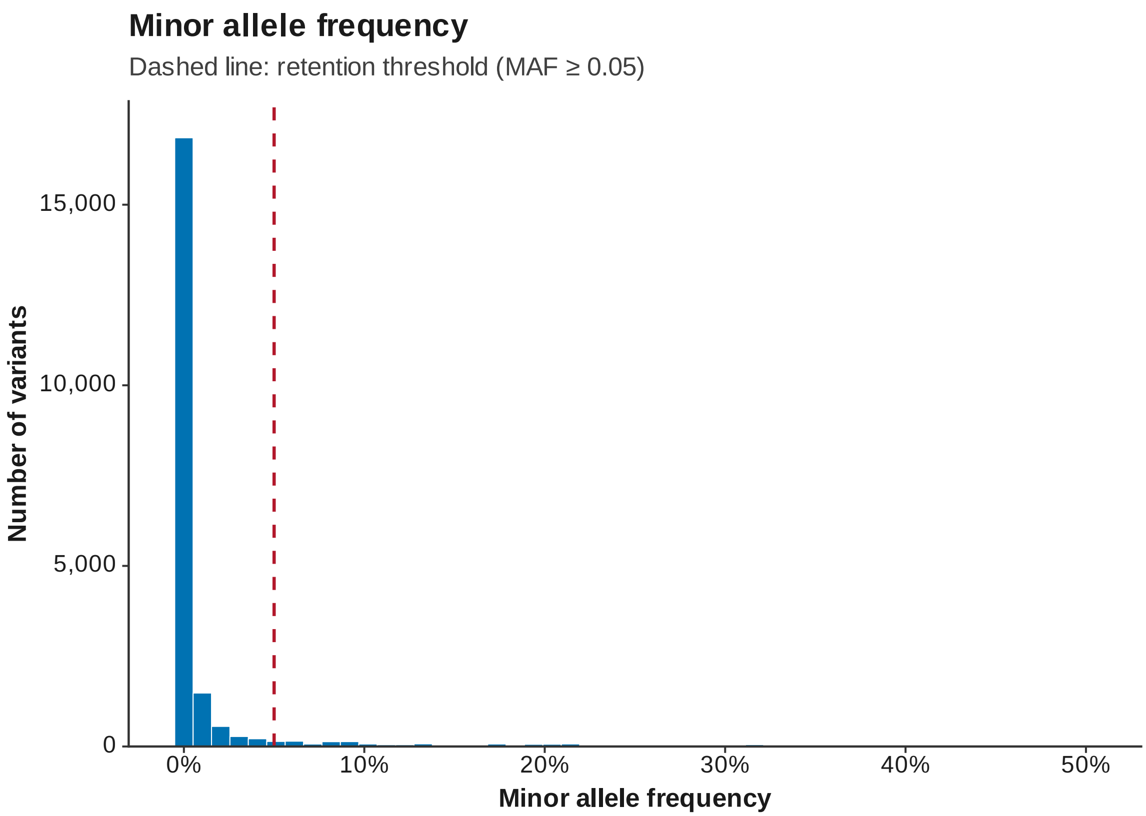

Figure 1: MAF

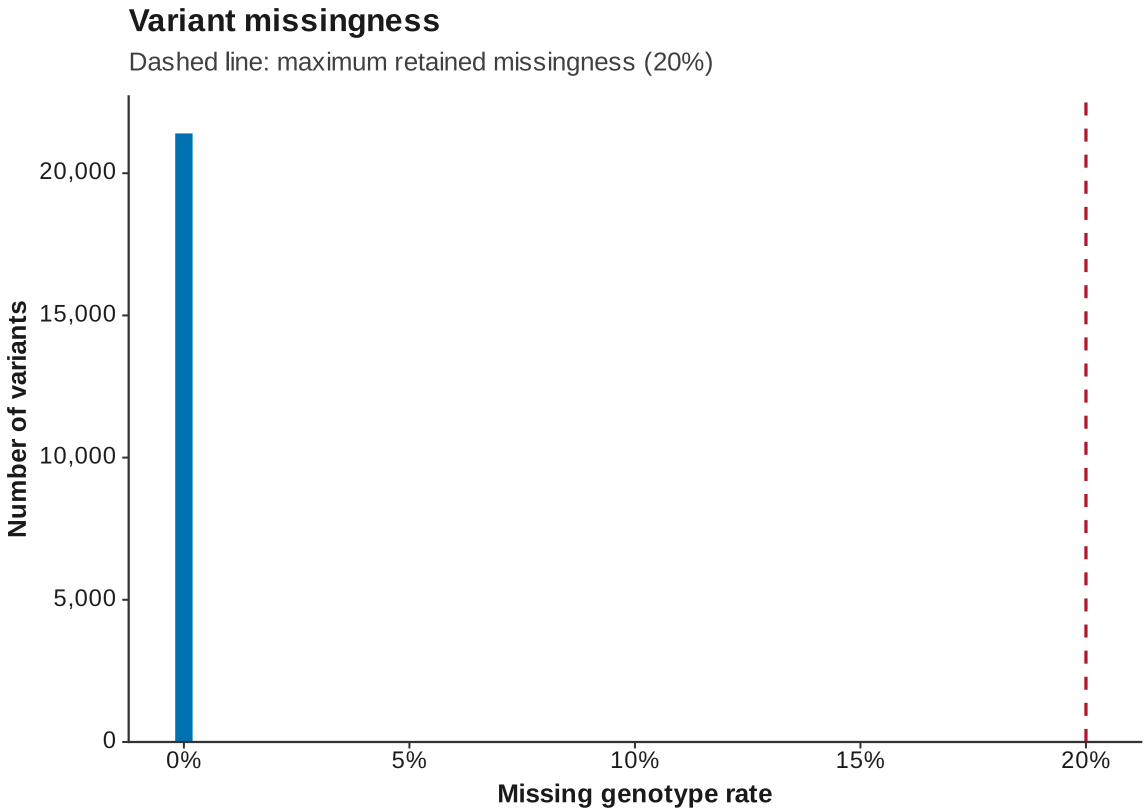

Figure 2: Variant missingness

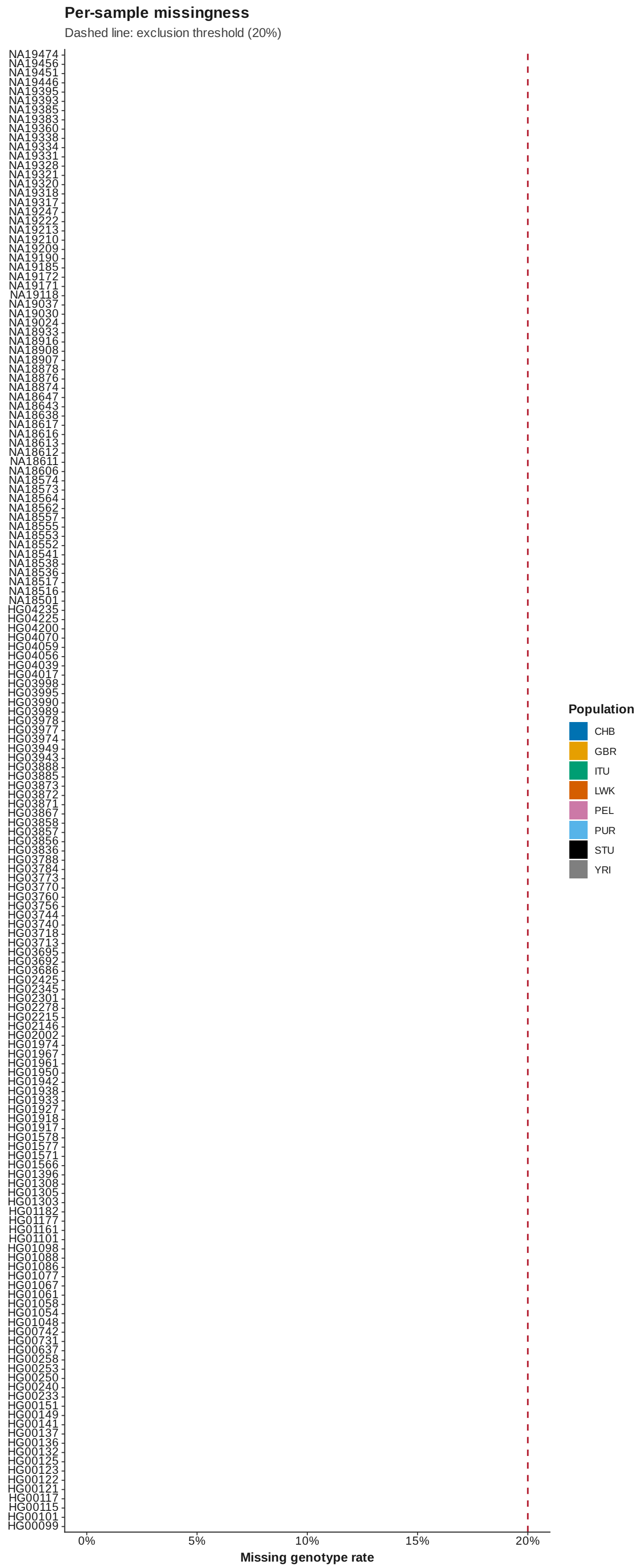

Figure 3: Sample missingness

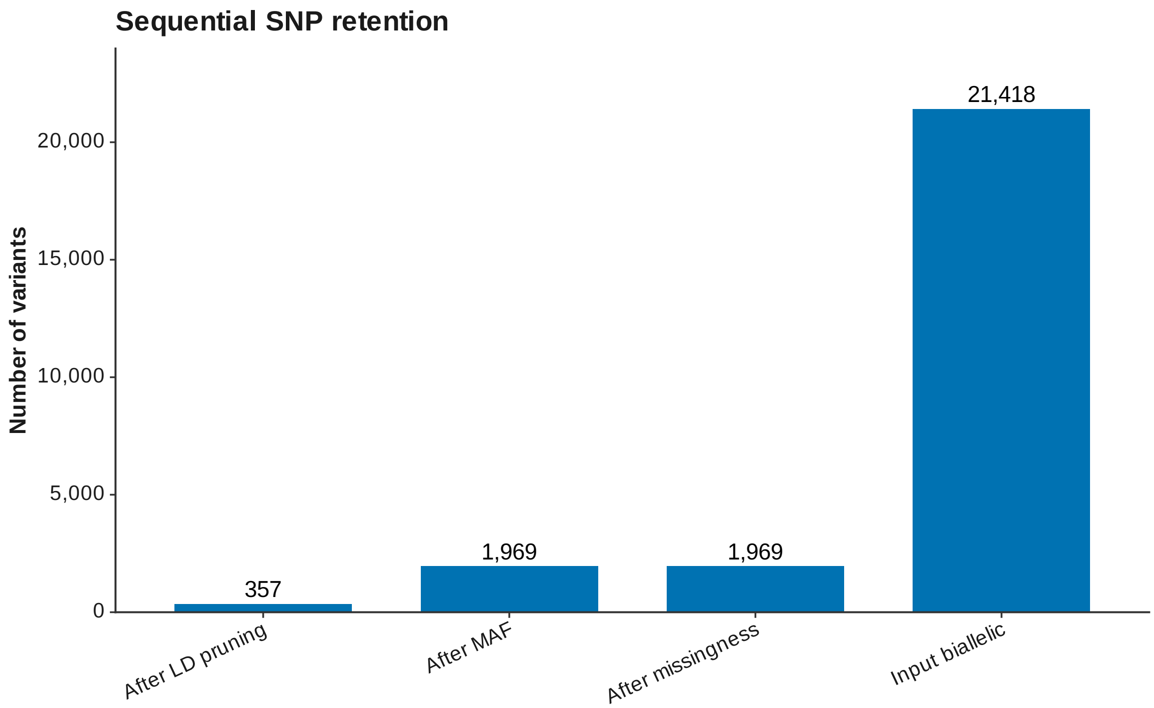

Figure 4: SNP retention

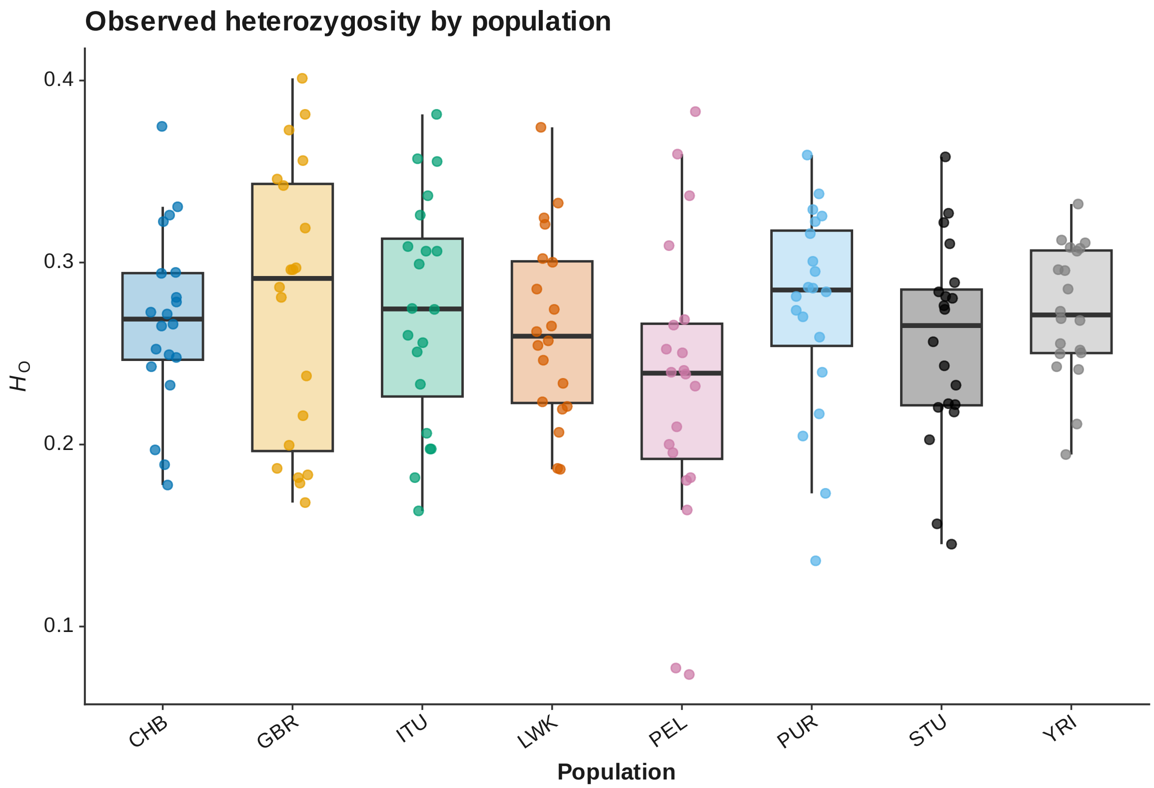

Figure 5: Sample heterozygosity

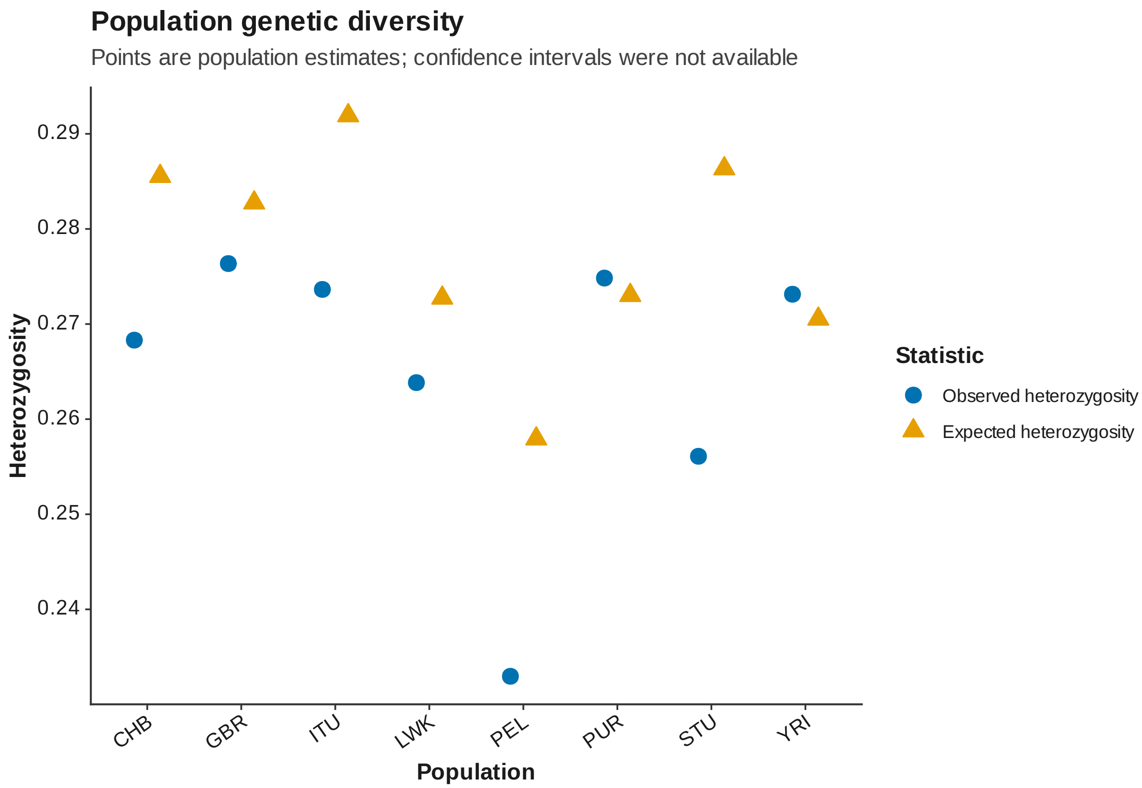

Figure 6: Population diversity

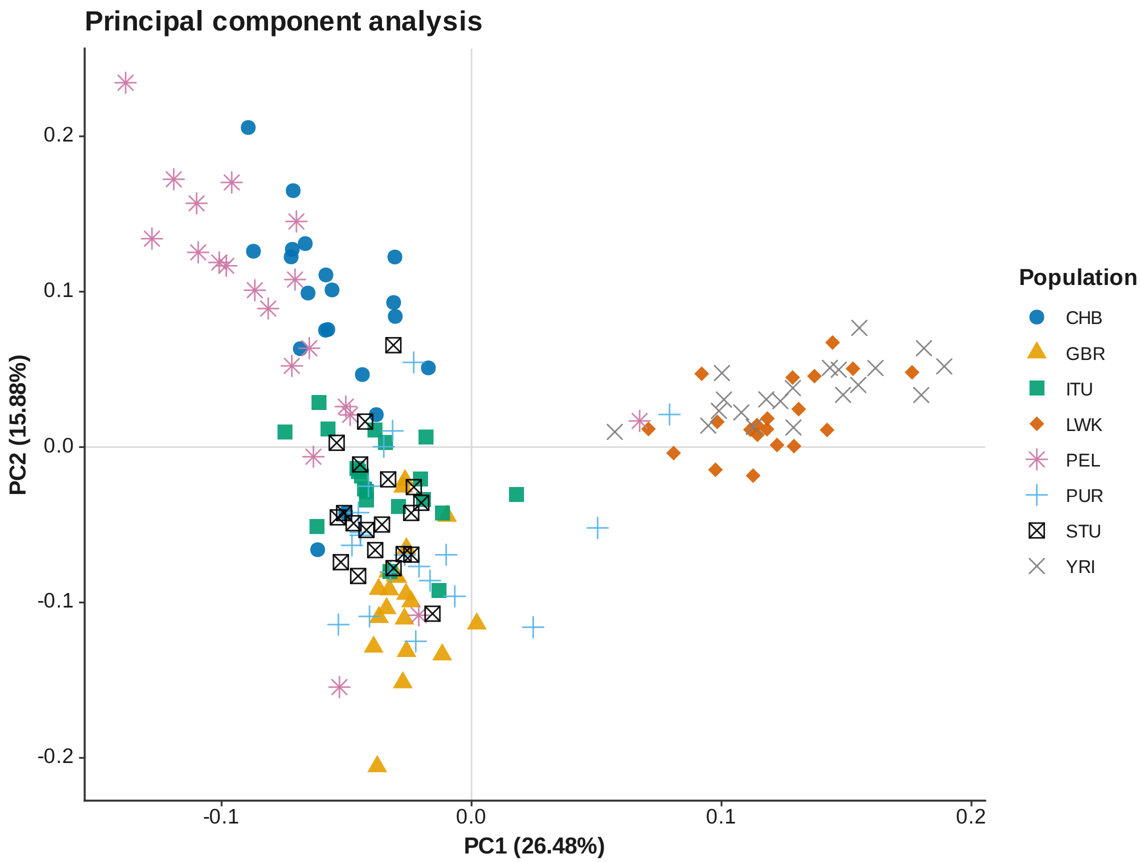

Figure 7: PCA PC 1 PC 2

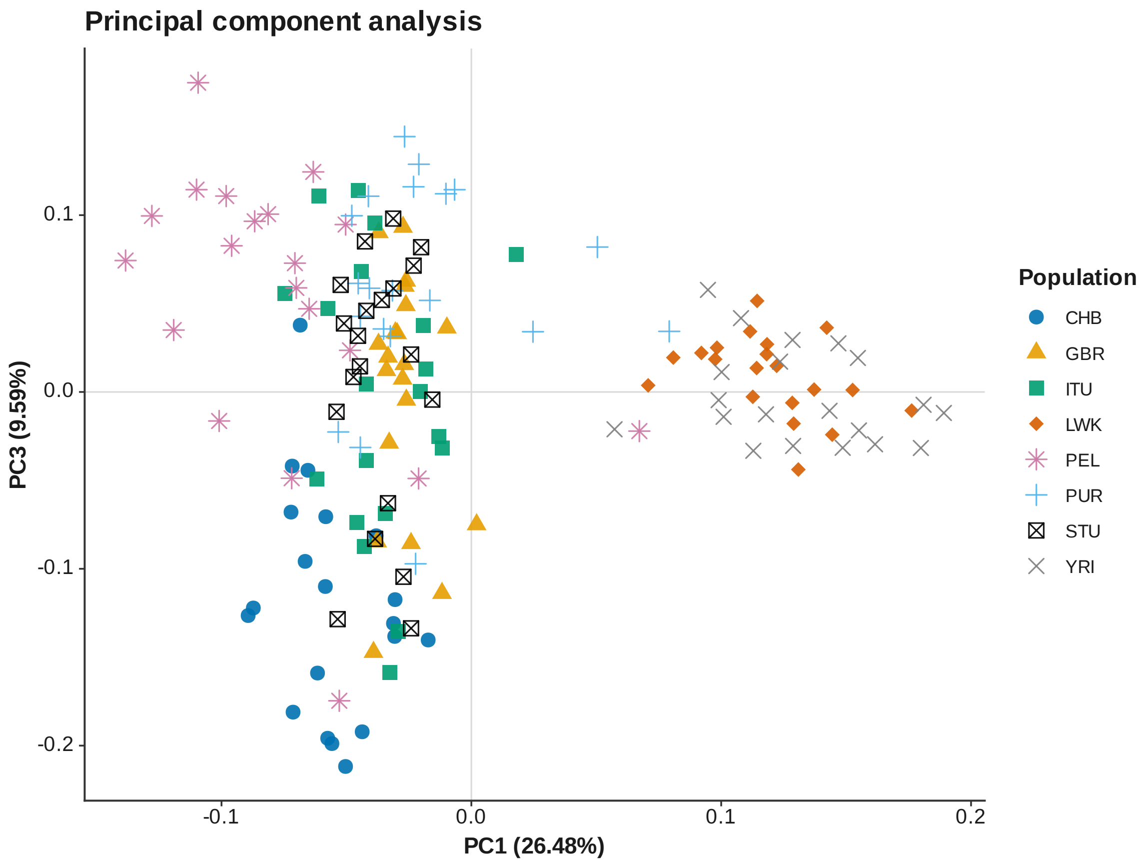

Figure 8: PCA PC 1 PC 3

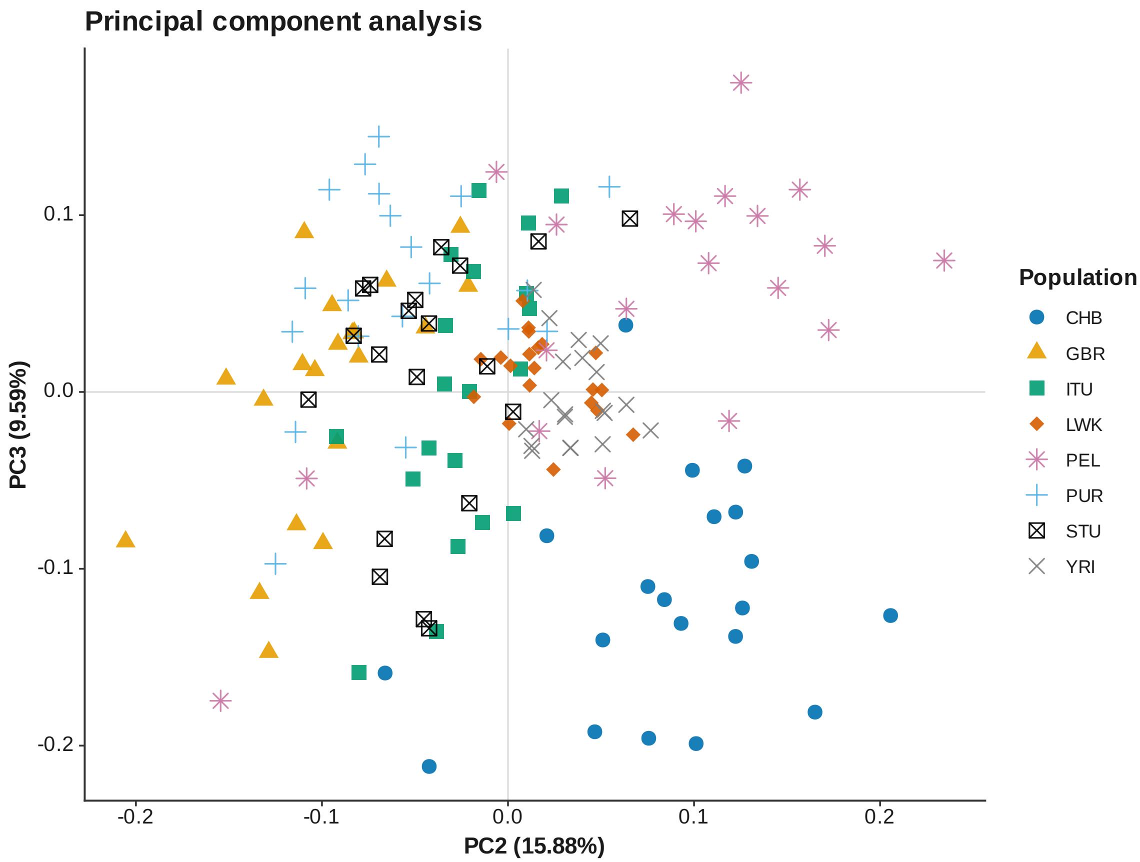

Figure 9: PCA PC 2 PC 3

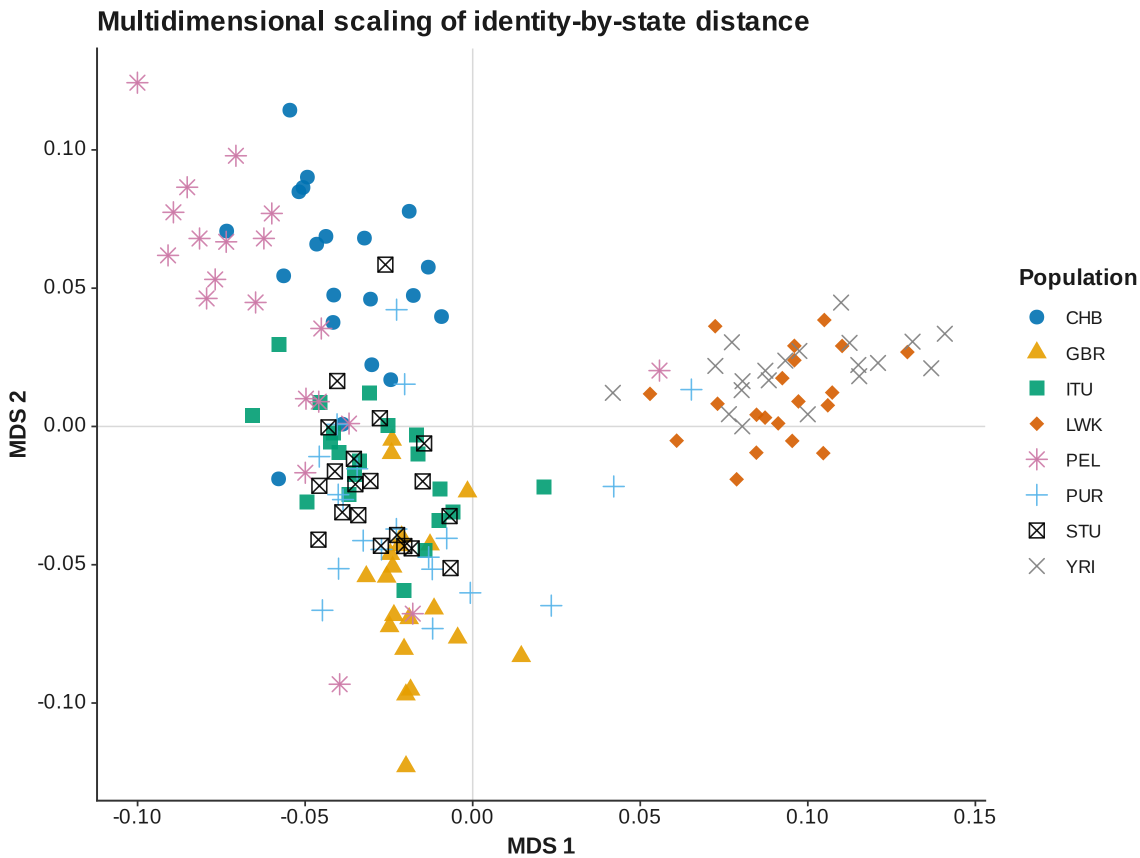

Figure 10: IBS MDS

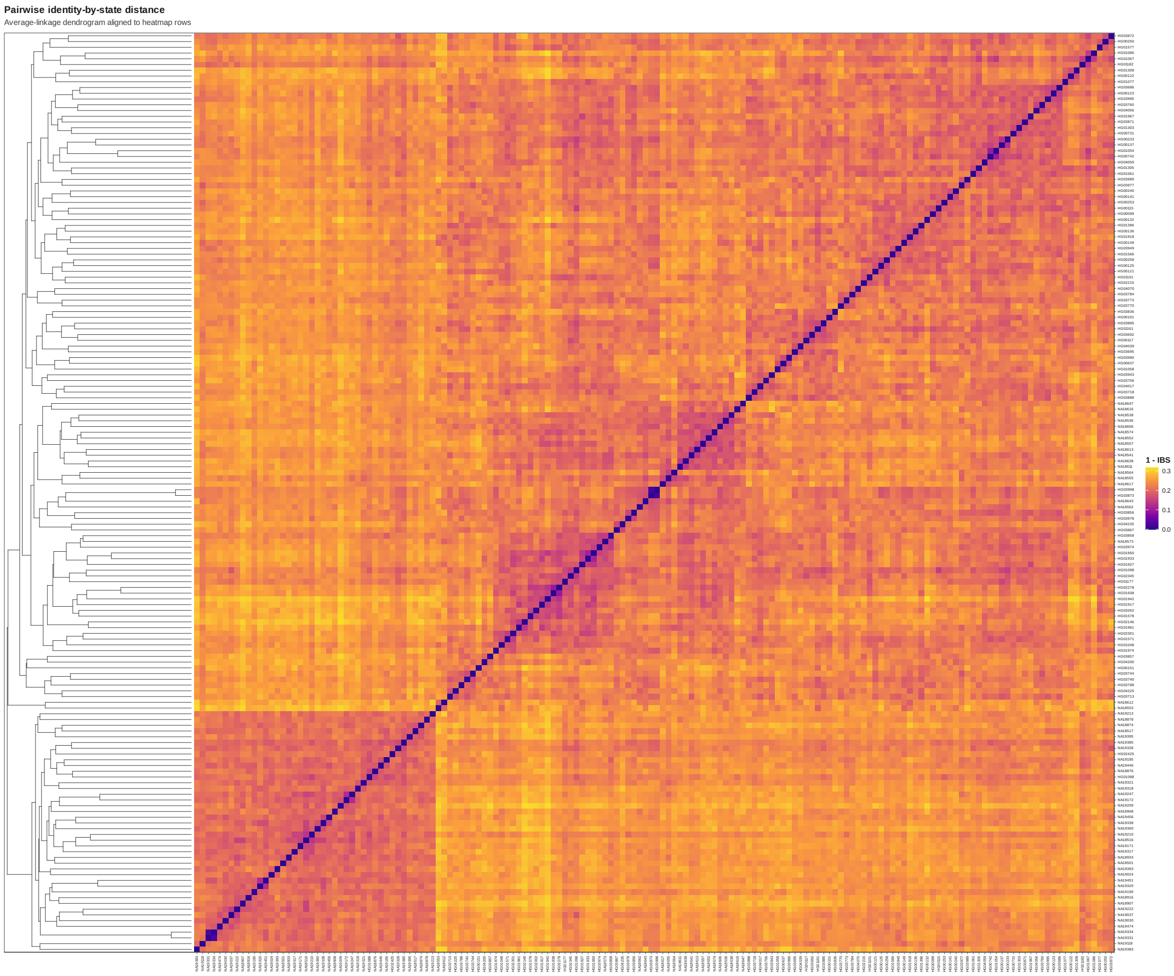

Figure 11: IBS heatmap

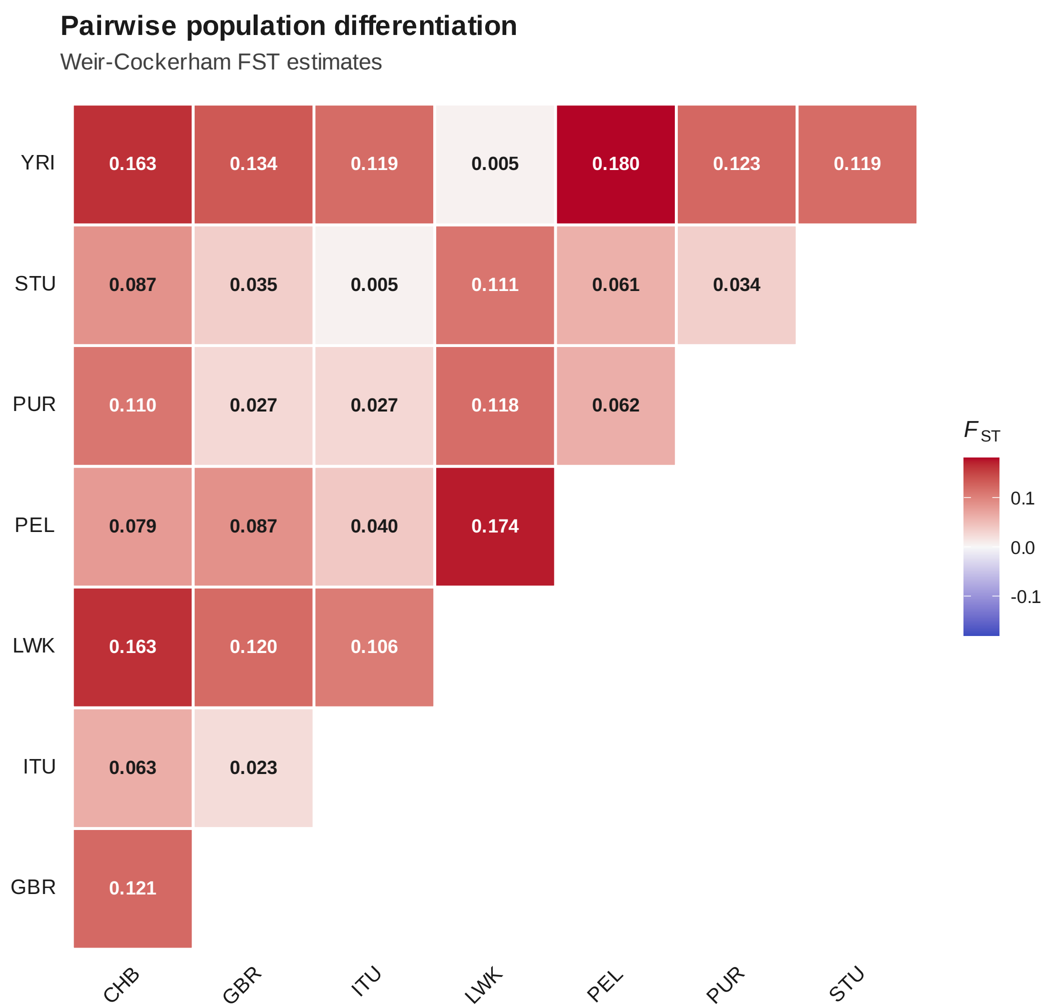

Figure 12: Pairwise FST

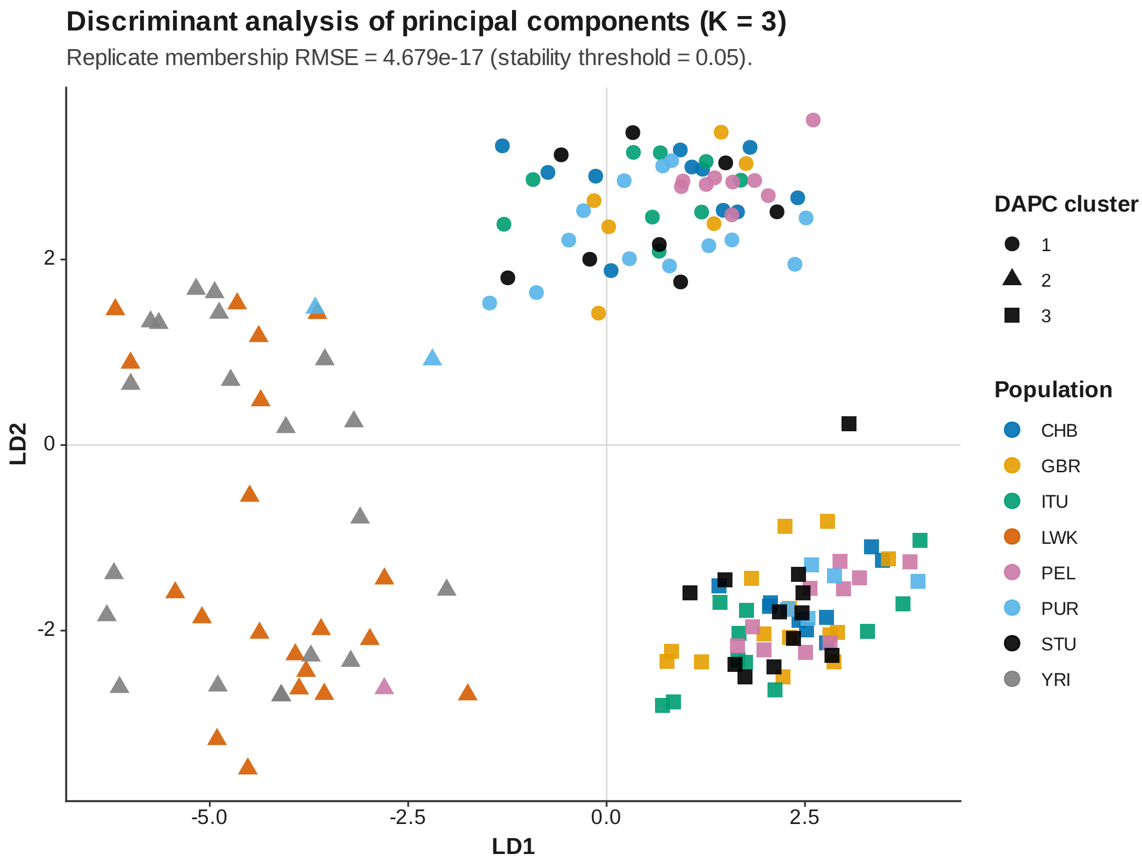

Figure 13: DAPC K 3

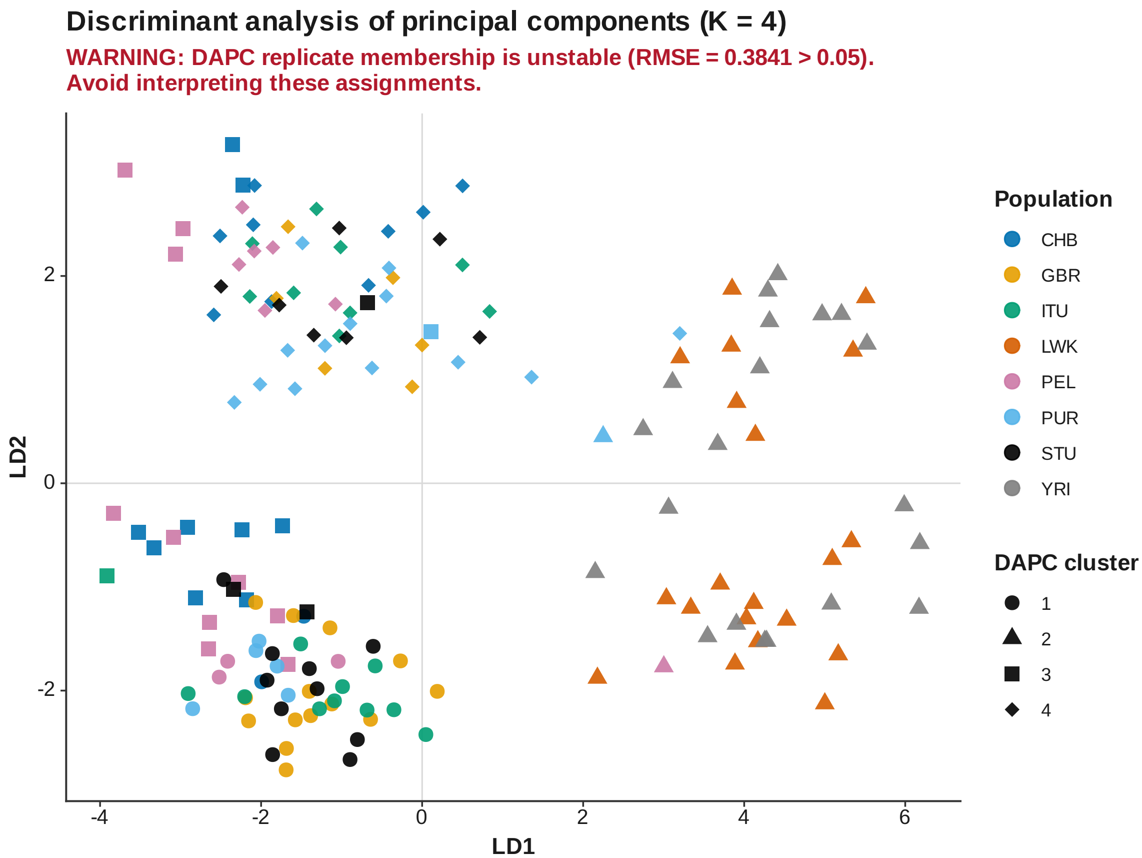

Figure 14: DAPC K 4

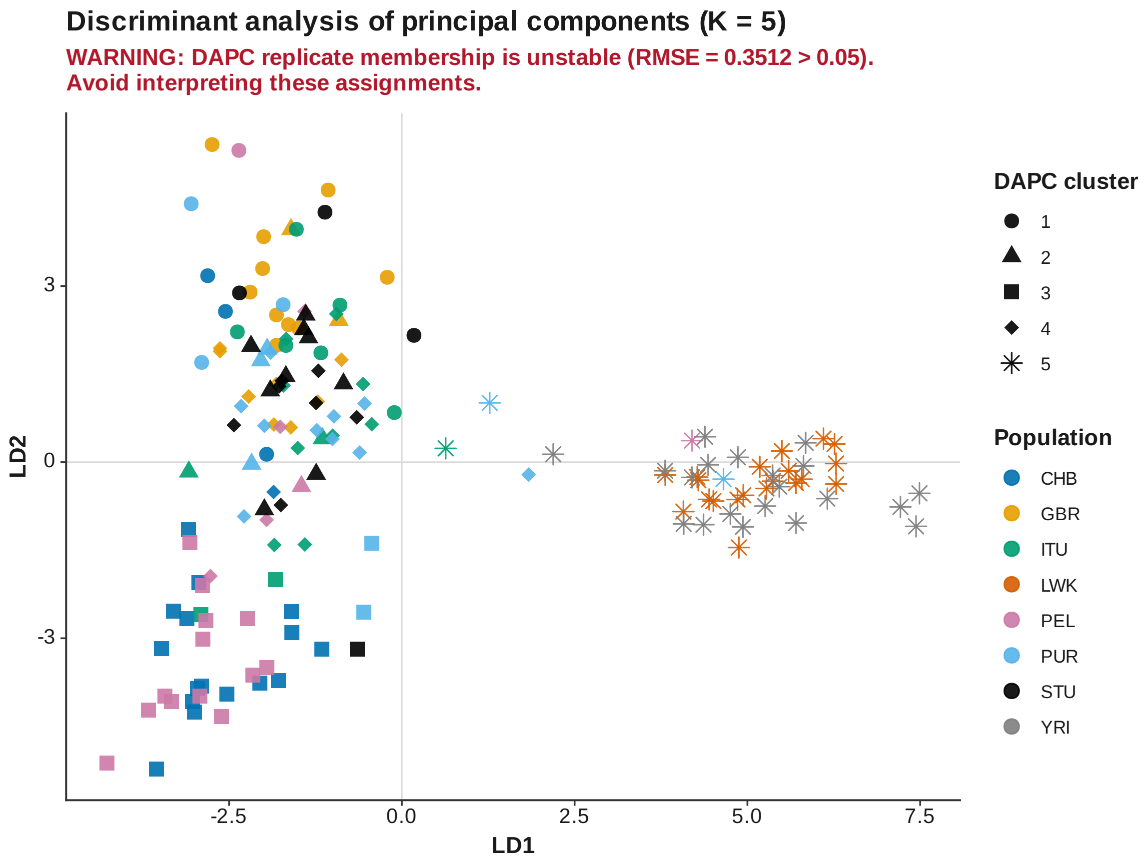

Figure 15: DAPC K 5

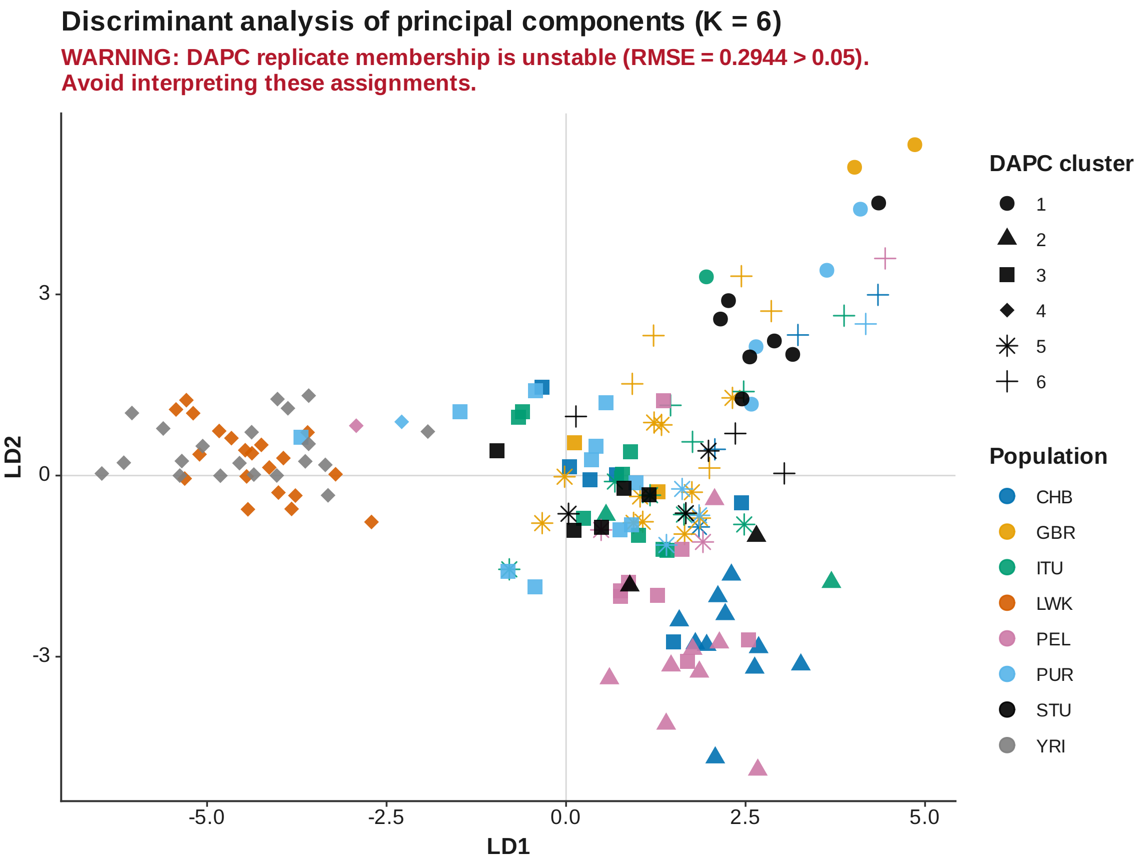

Figure 16: DAPC K 6

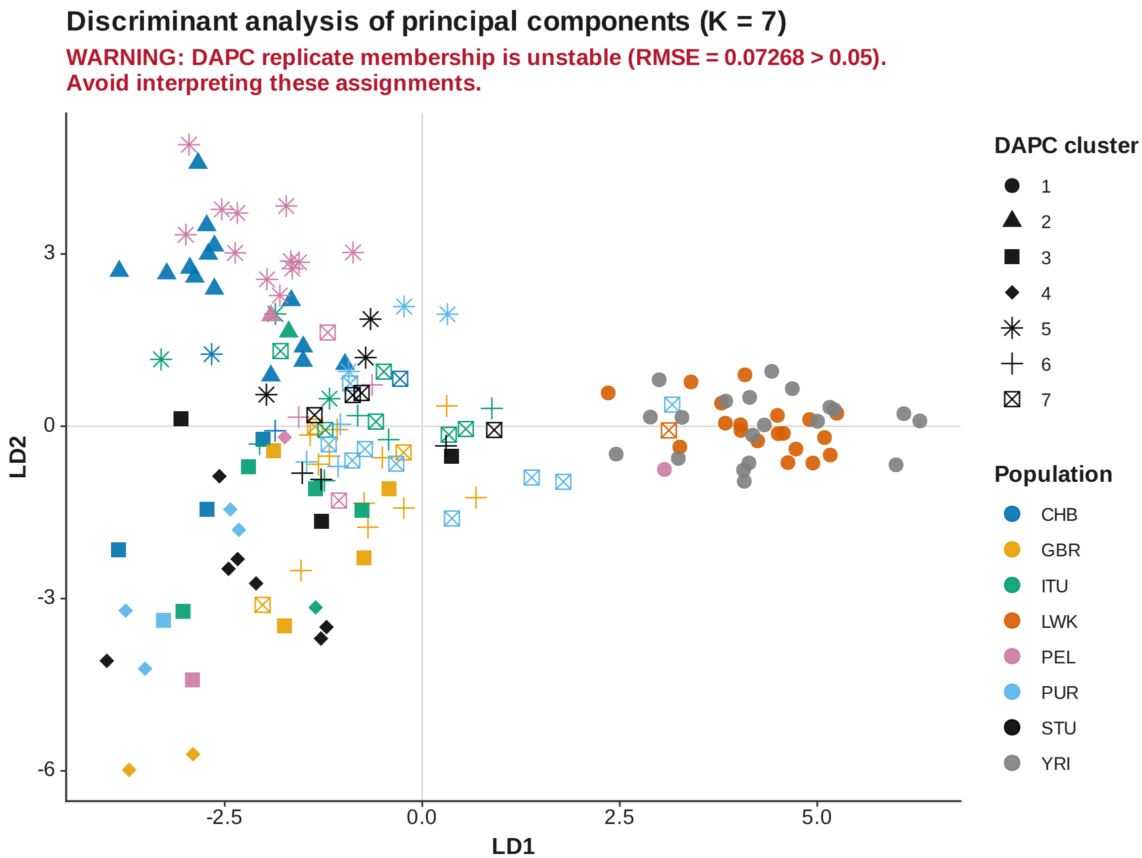

Figure 17: DAPC K 7

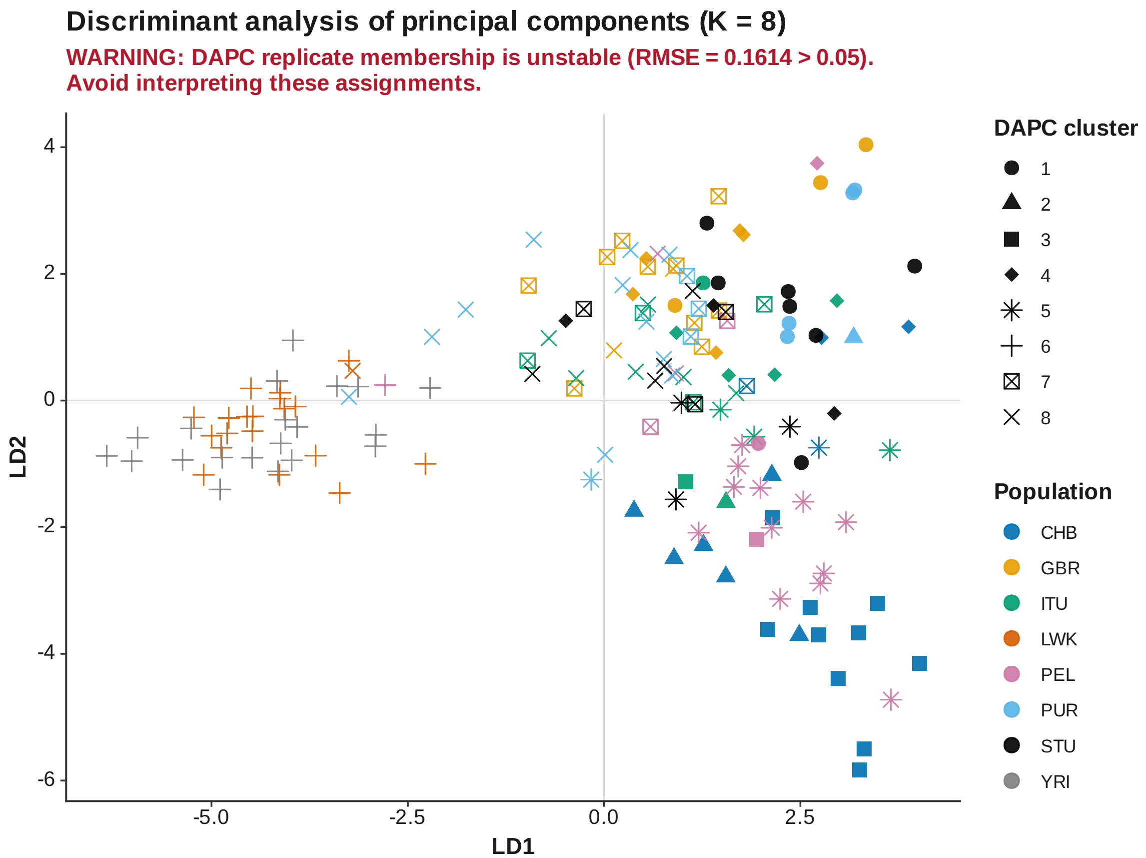

Figure 18: DAPC K 8

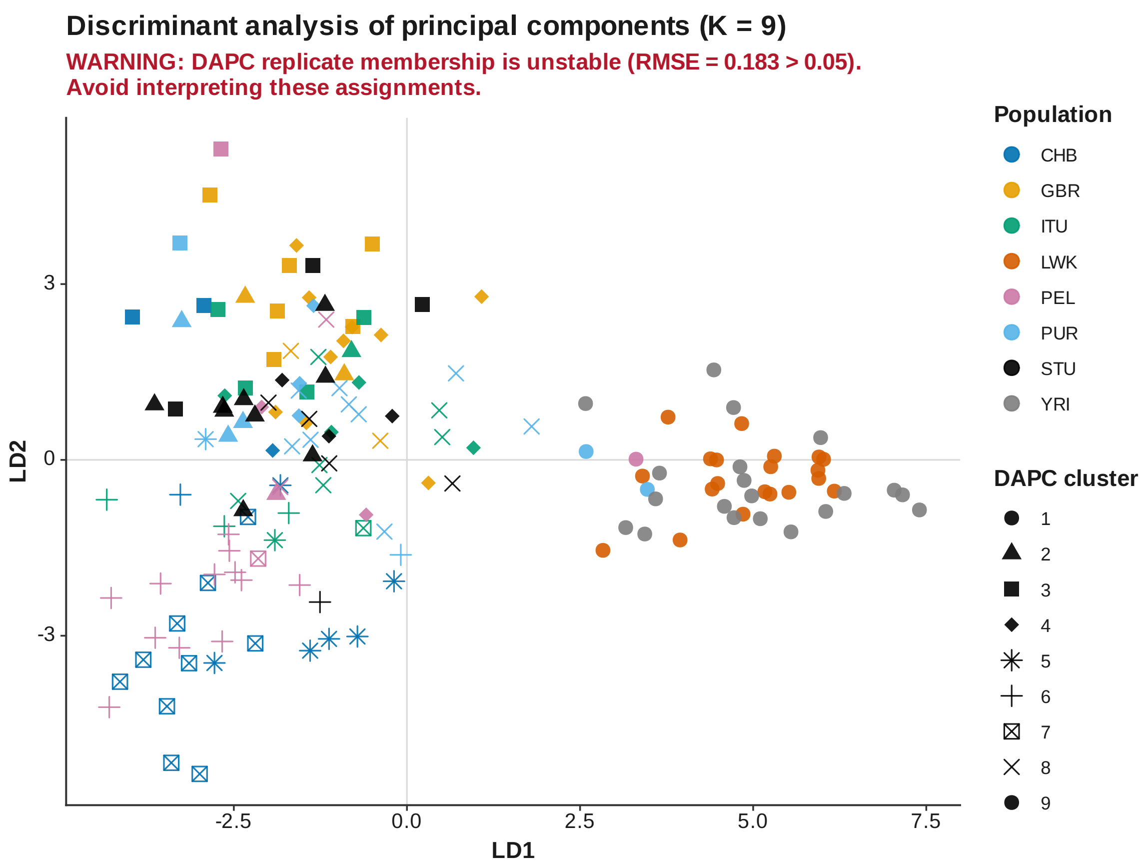

Figure 19: DAPC K 9

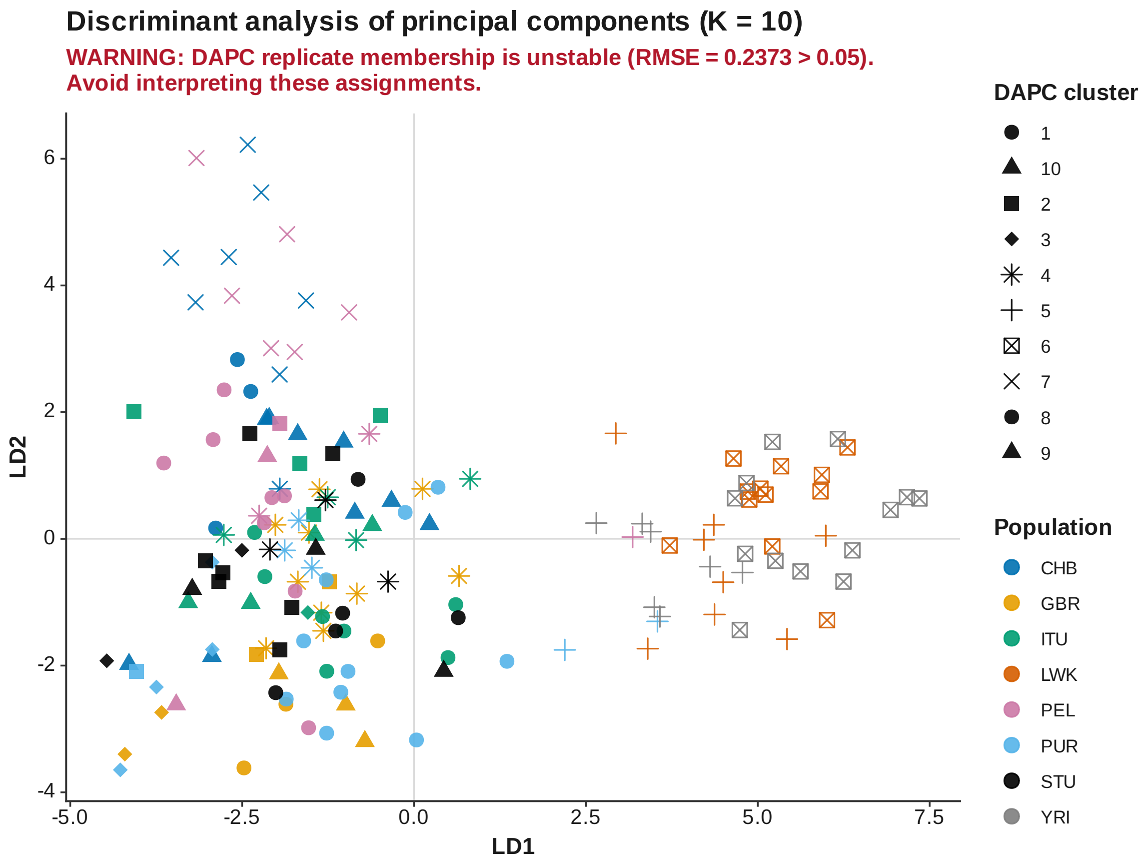

Figure 20: DAPC K 10

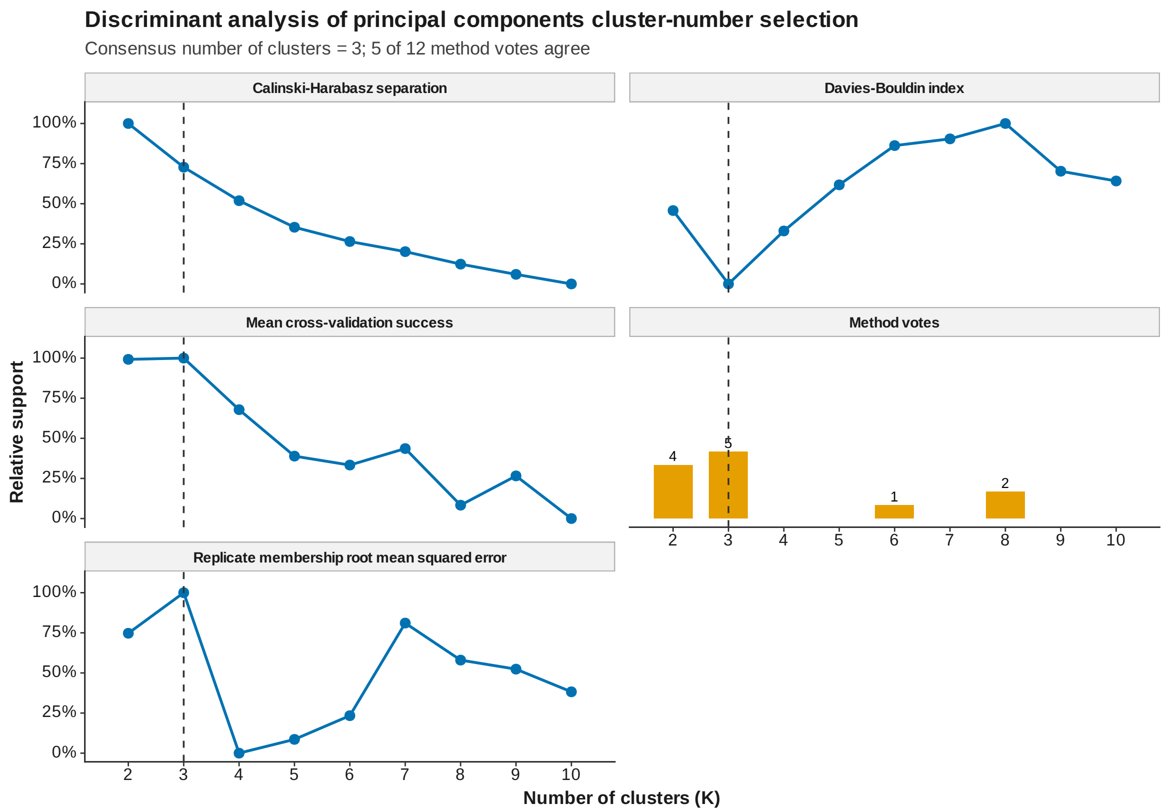

Figure 21: DAPC cluster number selection

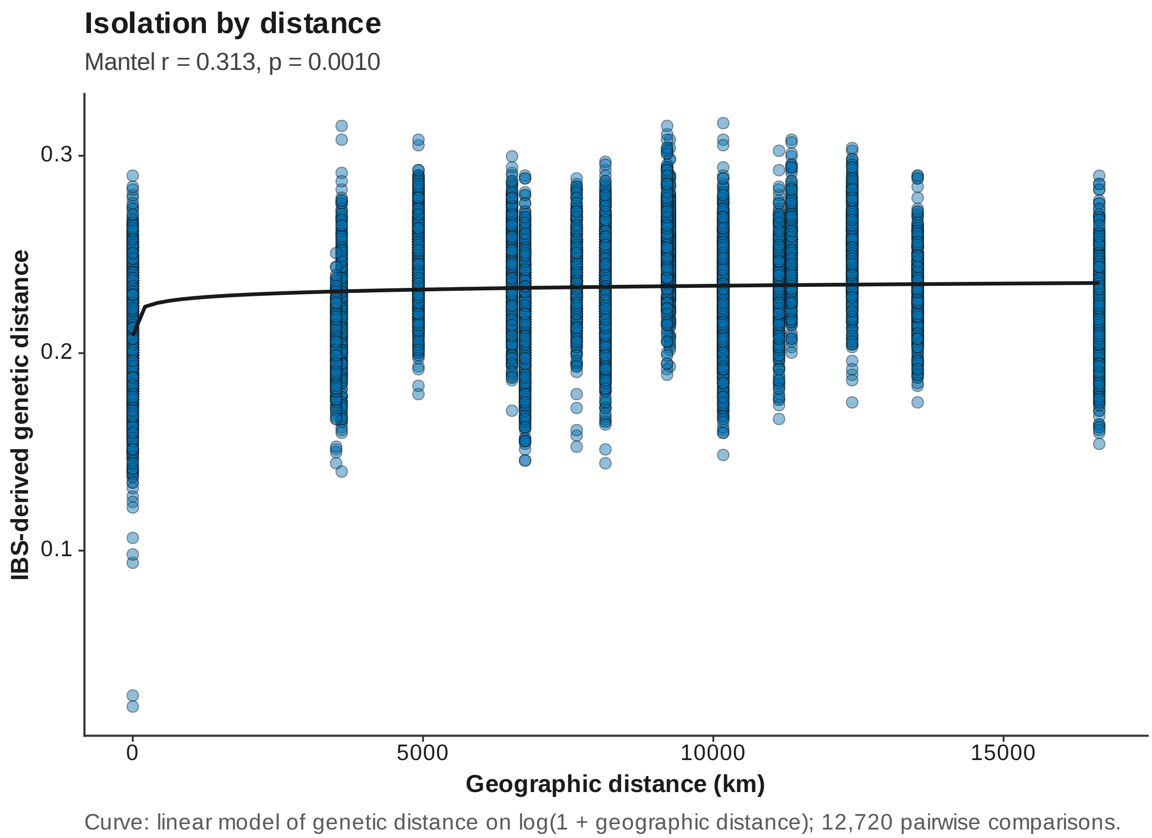

Figure 22: Isolation by distance

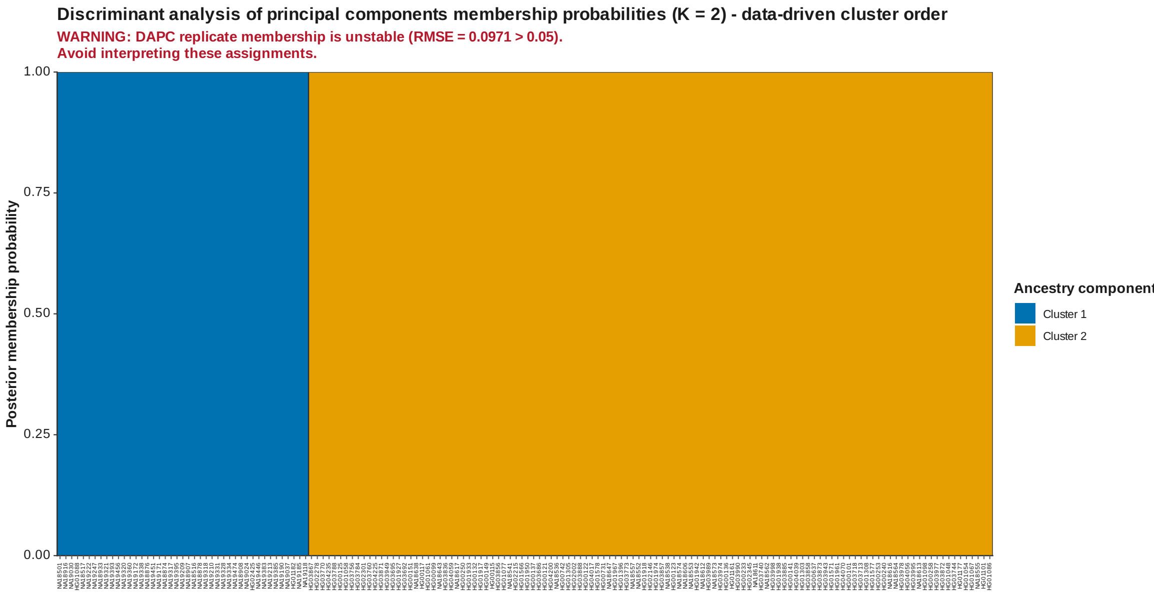

Figure 23: DAPC membership data driven K 2

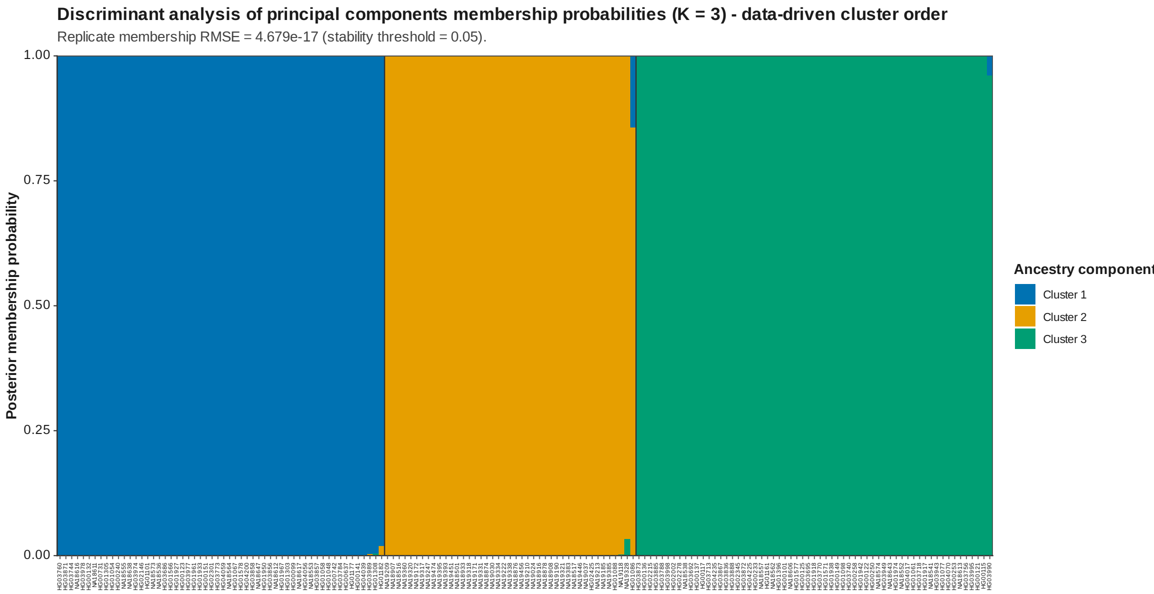

Figure 24: DAPC membership data driven K 3

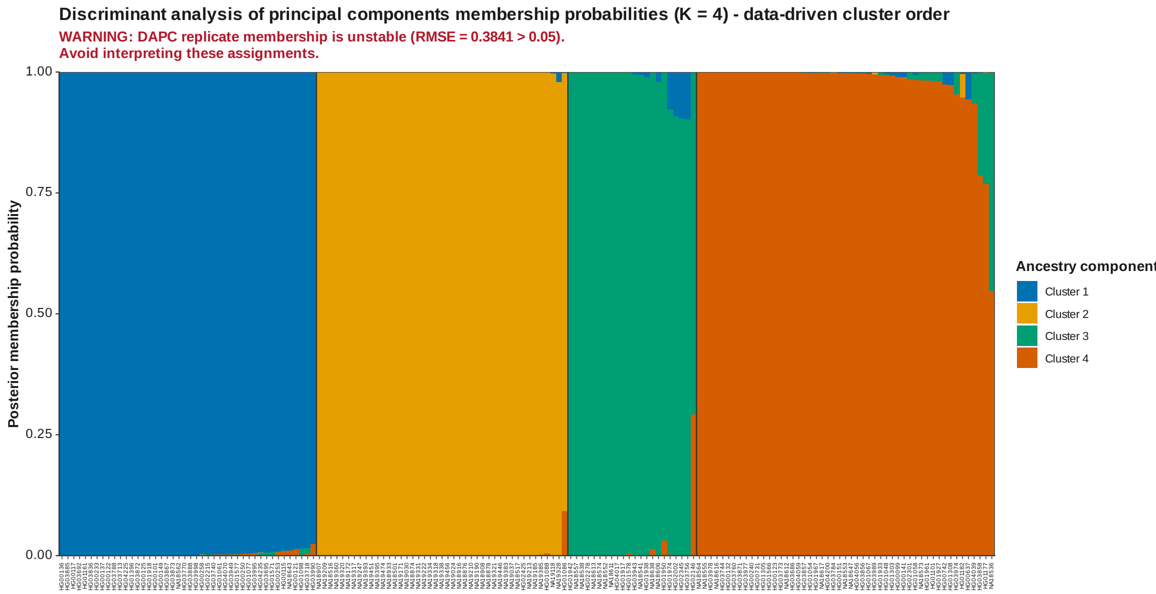

Figure 25: DAPC membership data driven K 4

Figure 26: DAPC membership data driven K 5

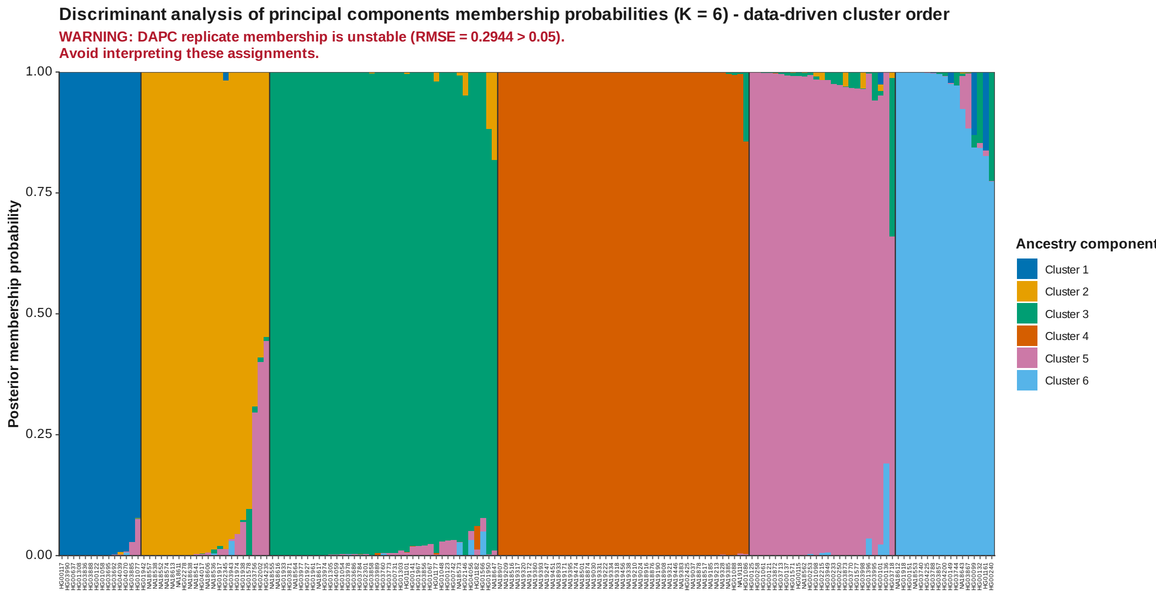

Figure 27: DAPC membership data driven K 6

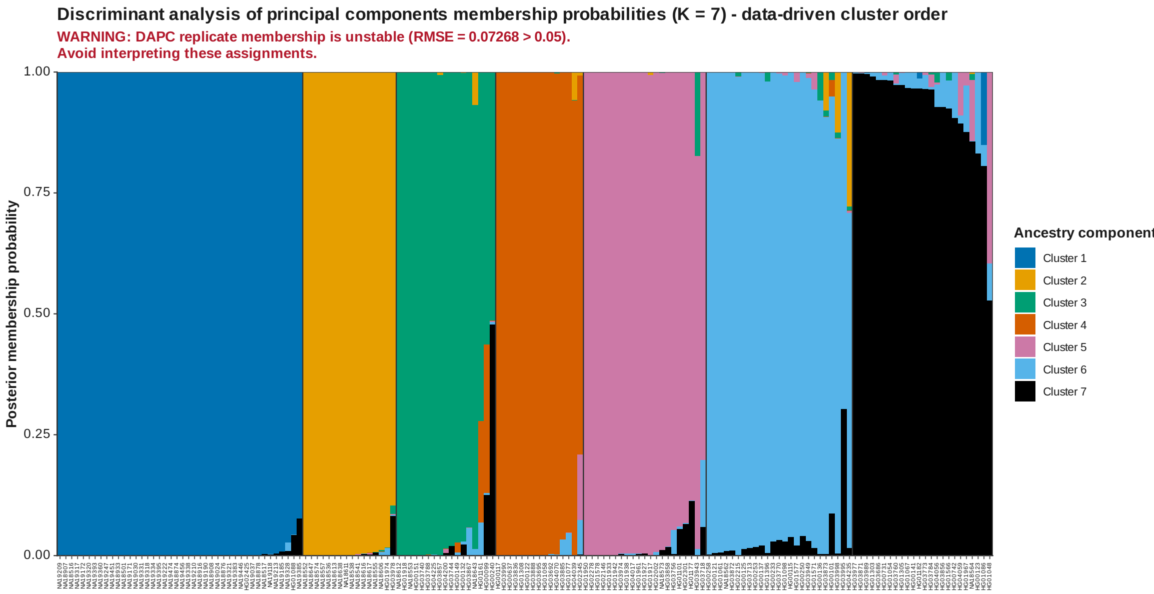

Figure 28: DAPC membership data driven K 7

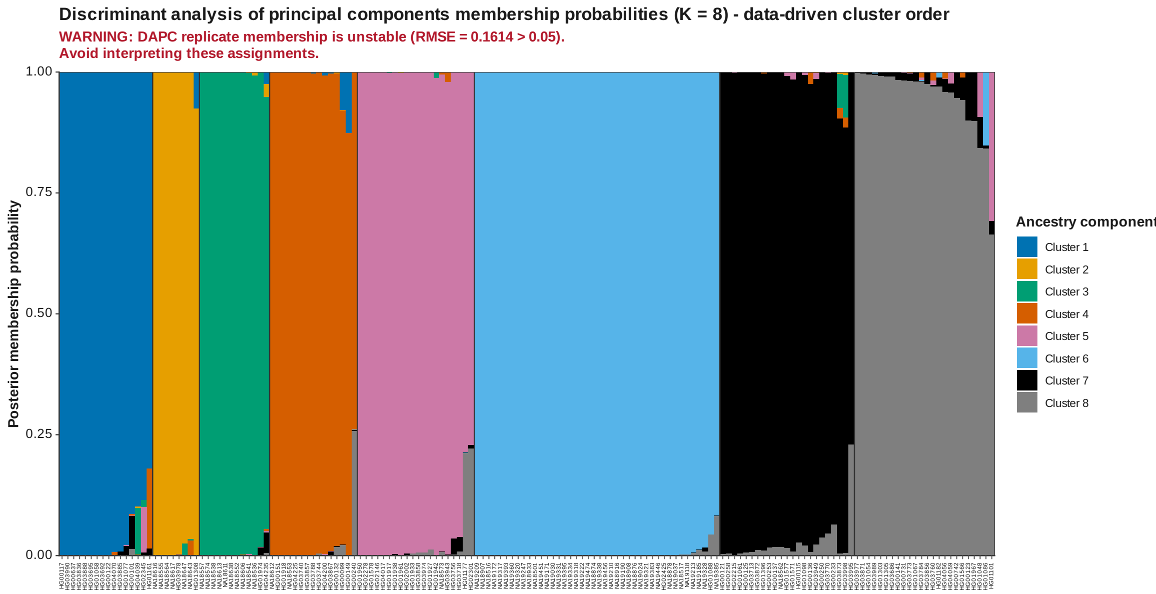

Figure 29: DAPC membership data driven K 8

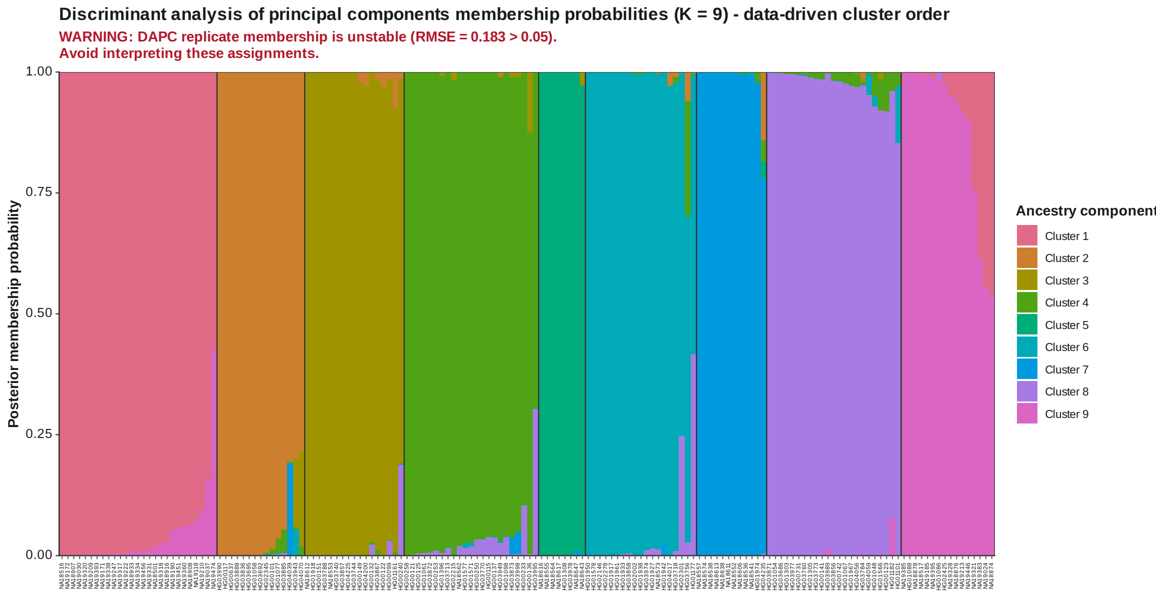

Figure 30: DAPC membership data driven K 9

Figure 31: DAPC membership data driven K 10

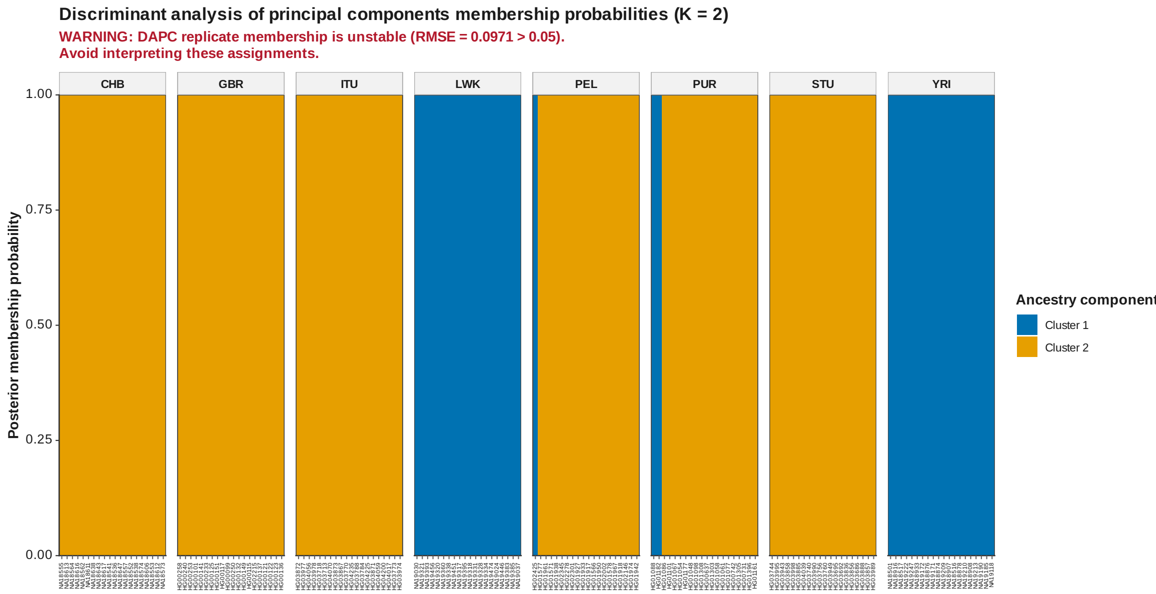

Figure 32: DAPC membership K 2

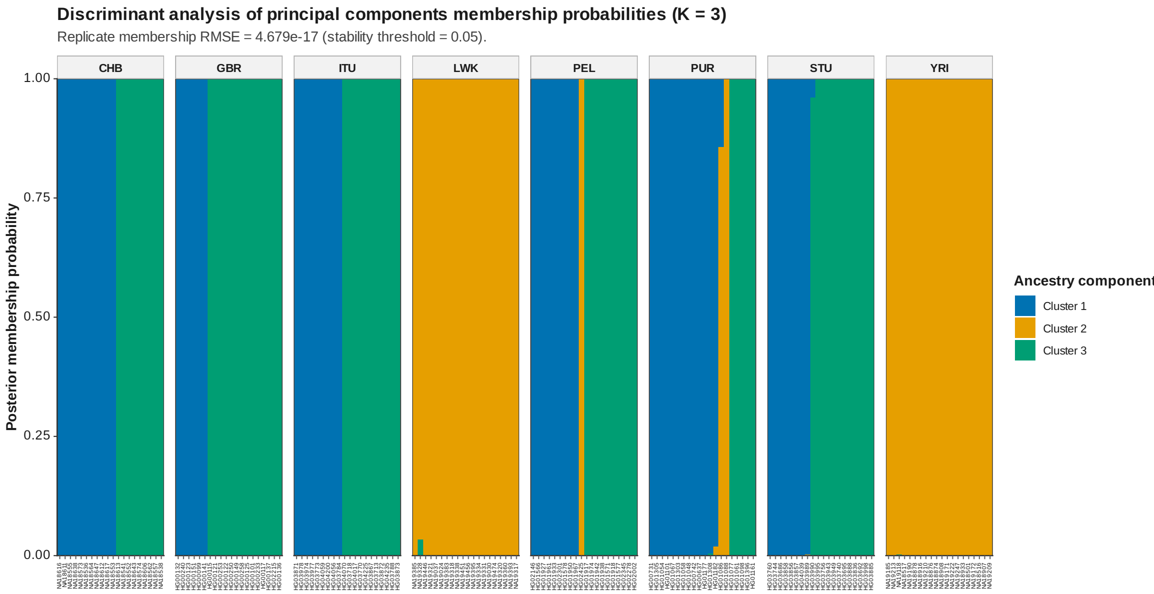

Figure 33: DAPC membership K 3

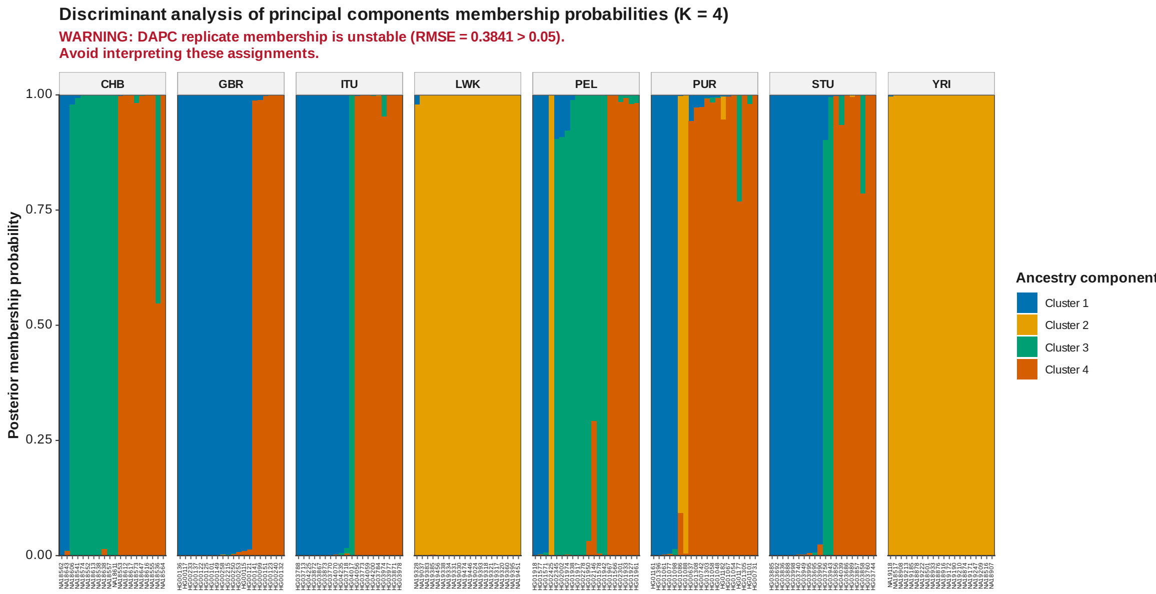

Figure 34: DAPC membership K 4

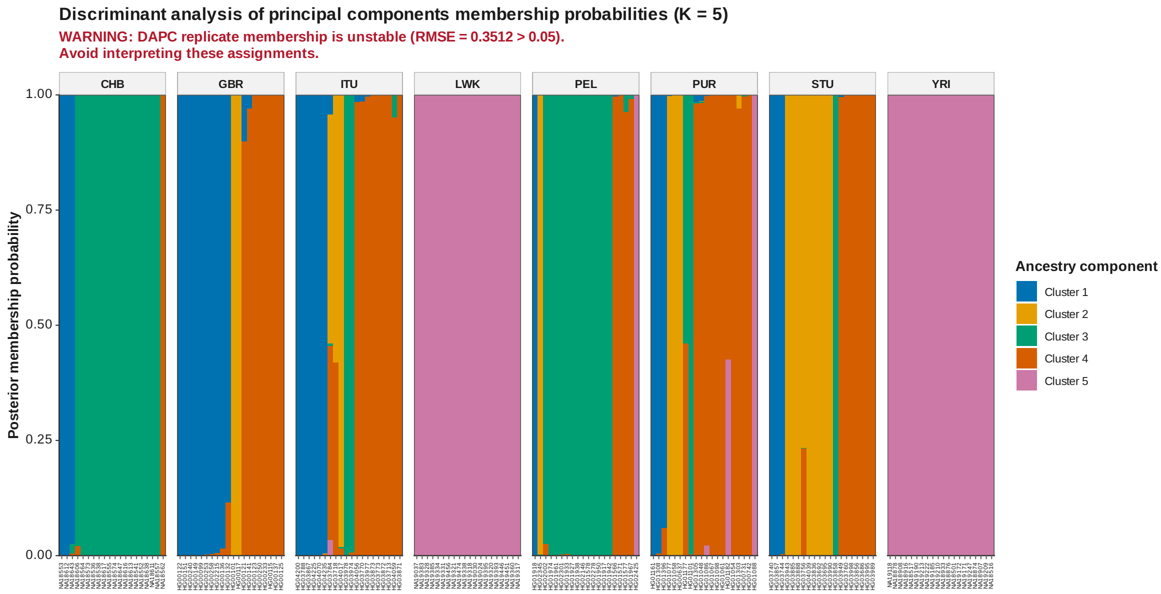

Figure 35: DAPC membership K 5

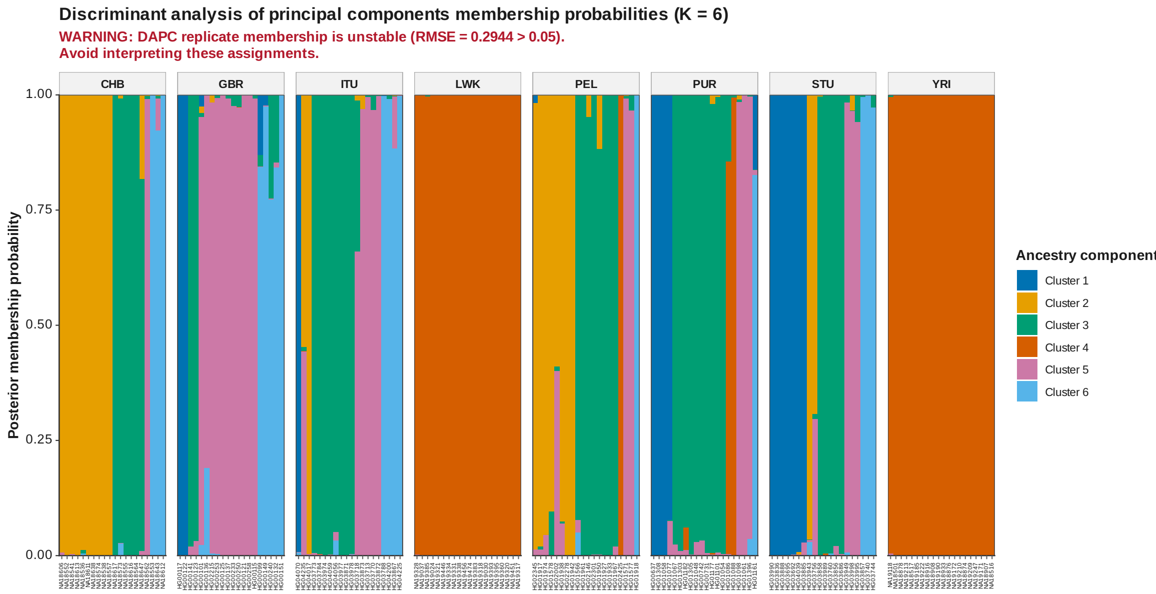

Figure 36: DAPC membership K 6

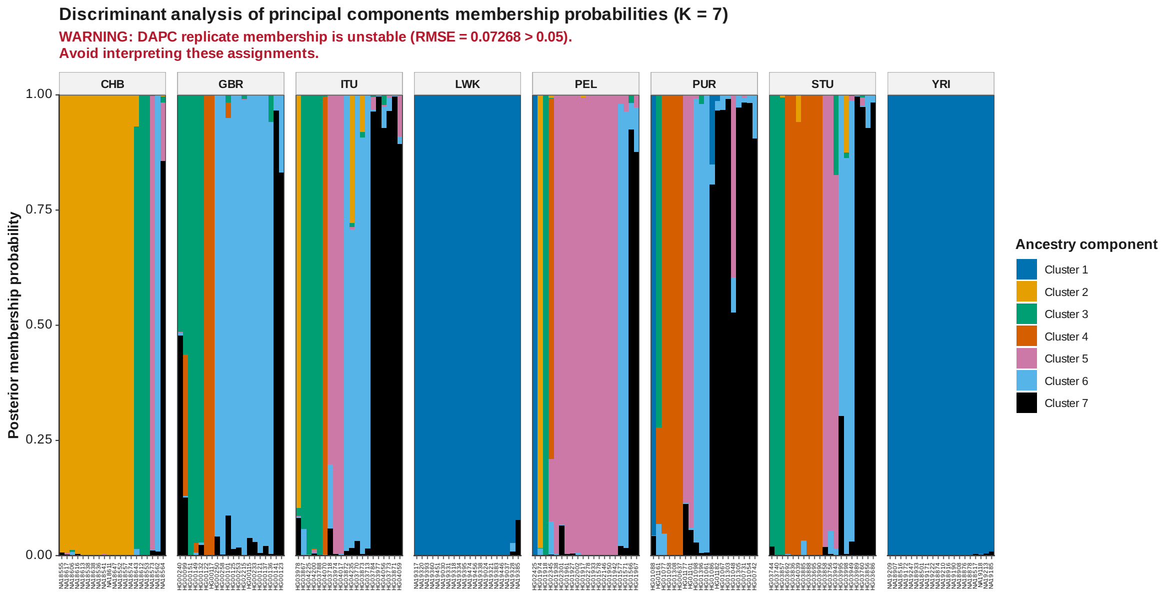

Figure 37: DAPC membership K 7

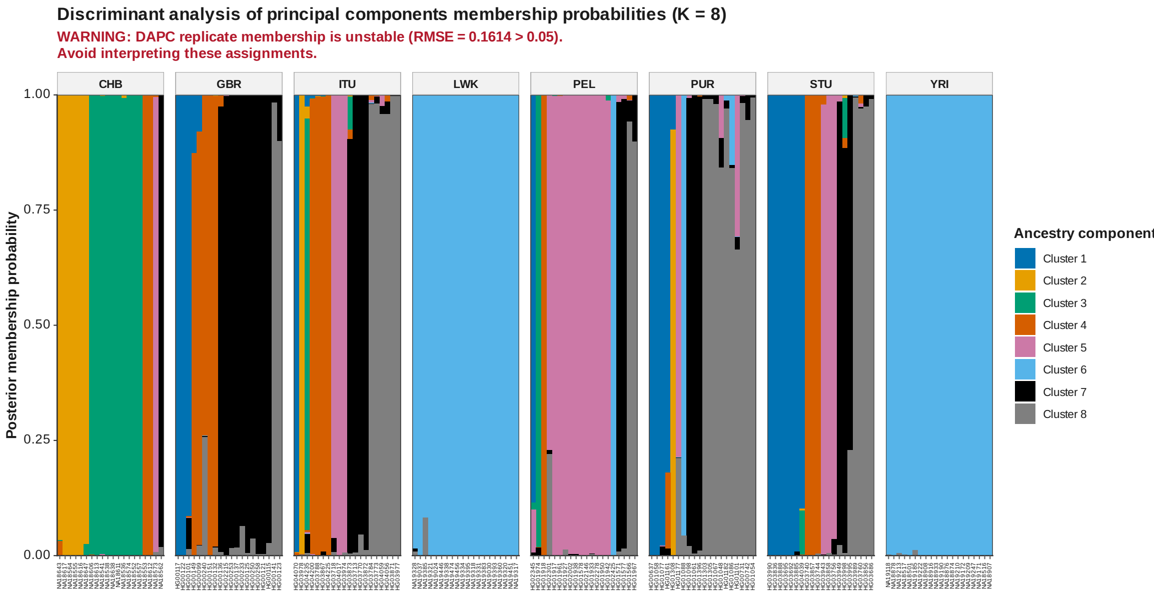

Figure 38: DAPC membership K 8

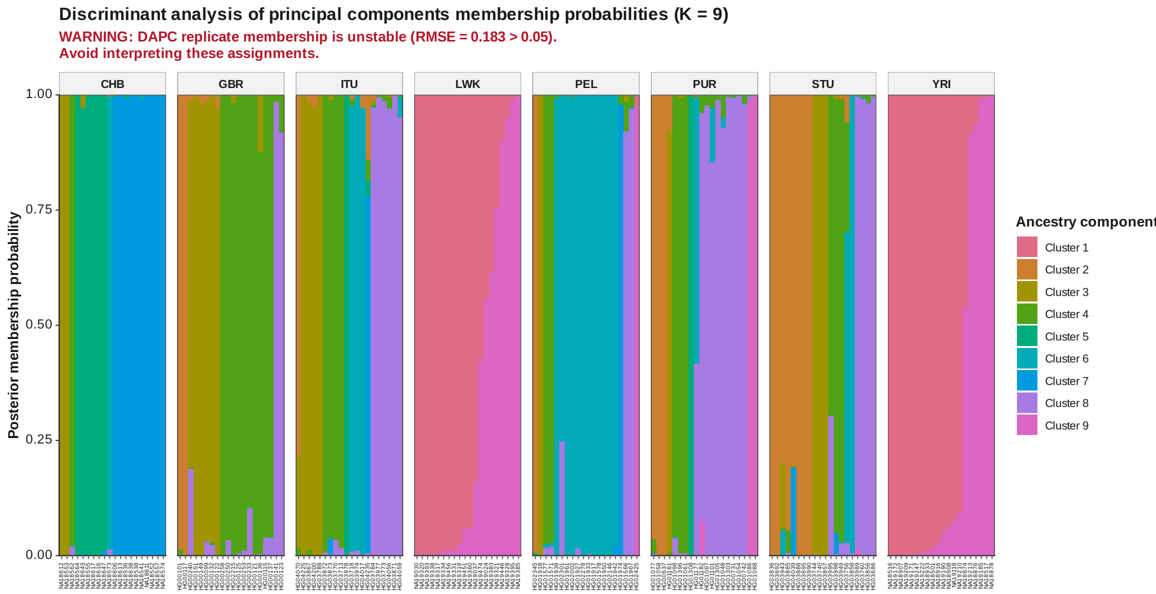

Figure 39: DAPC membership K 9

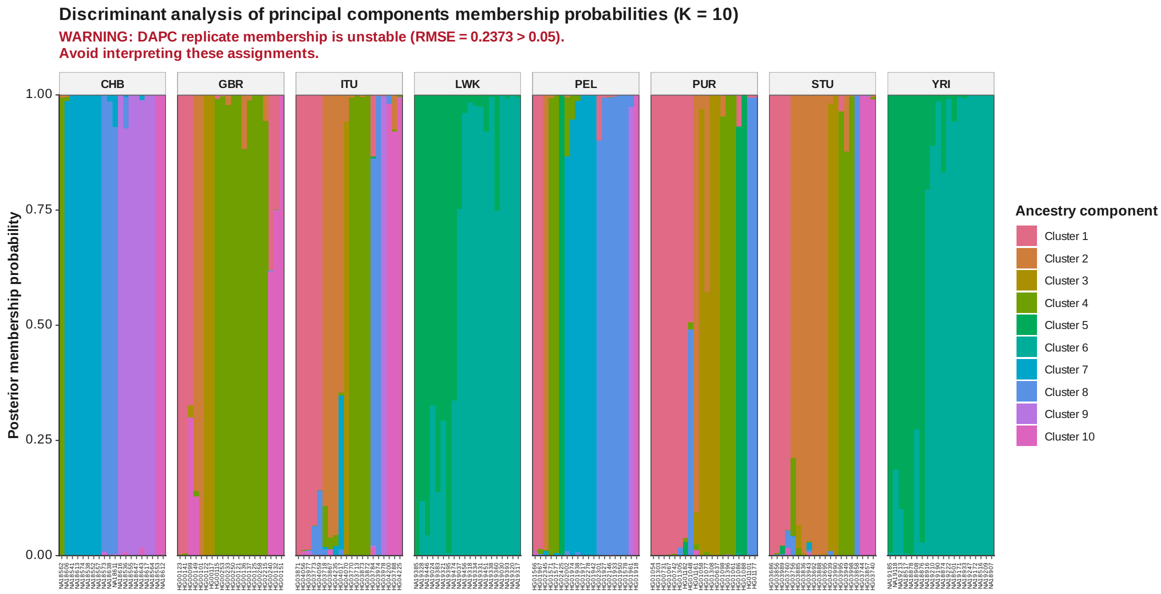

Figure 40: DAPC membership K 10

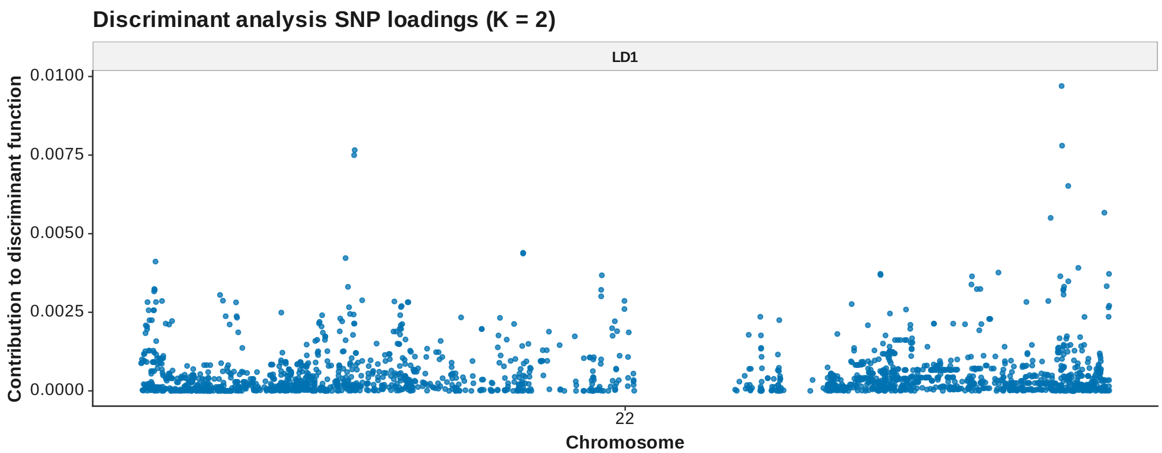

Figure 41: DAPC loadings manhattan K 2

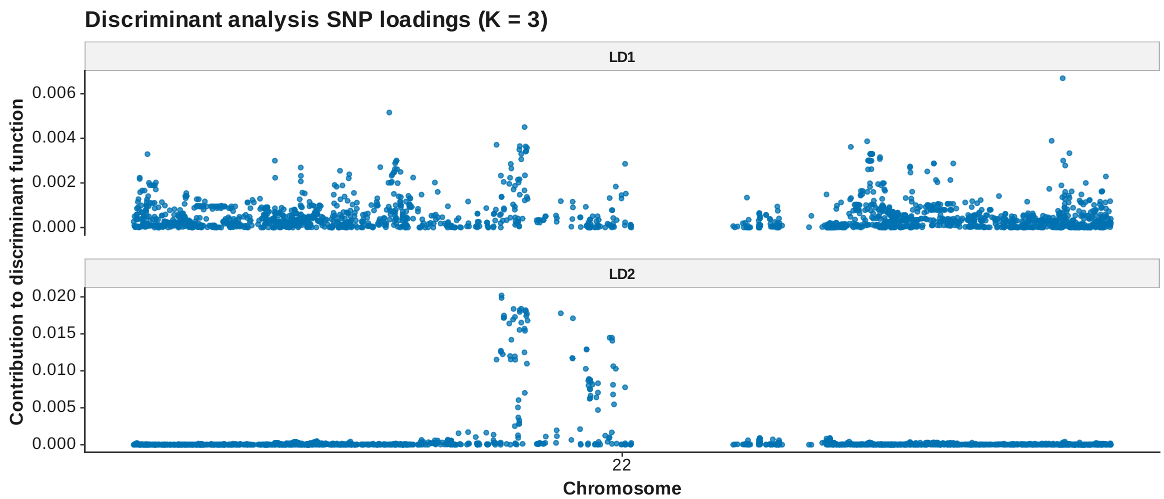

Figure 42: DAPC loadings manhattan K 3

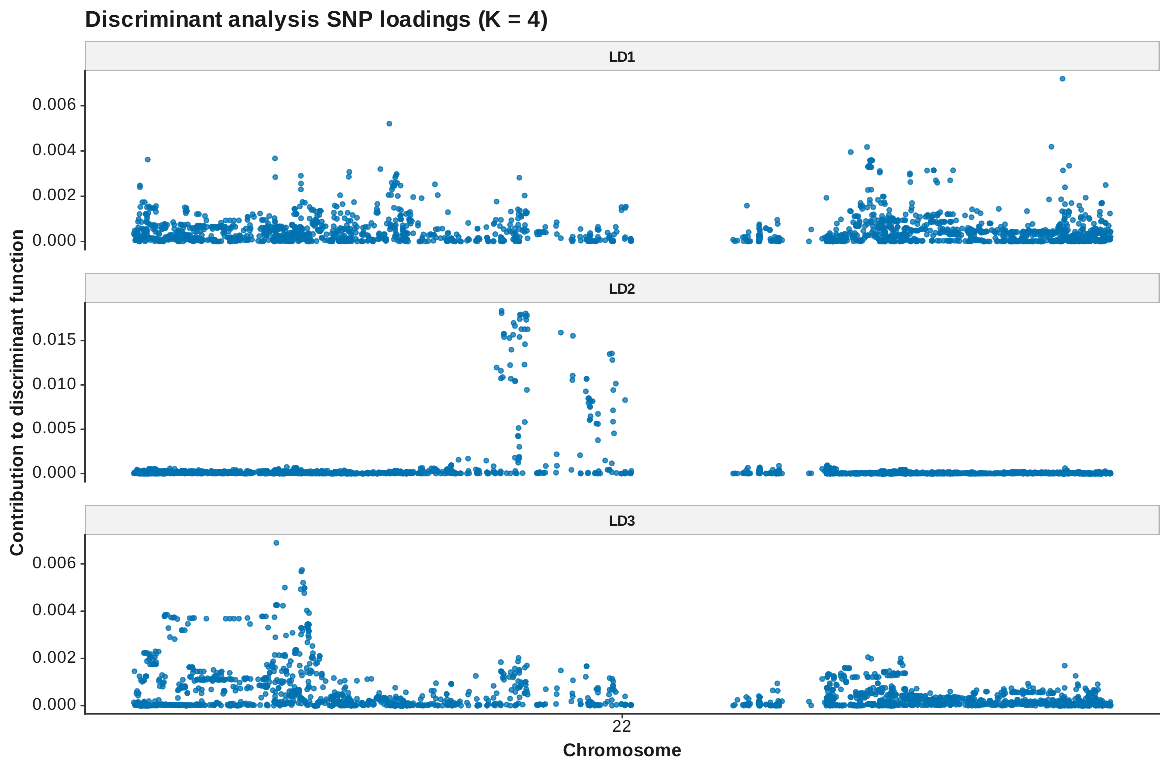

Figure 43: DAPC loadings manhattan K 4

Figure 44: DAPC loadings manhattan K 5

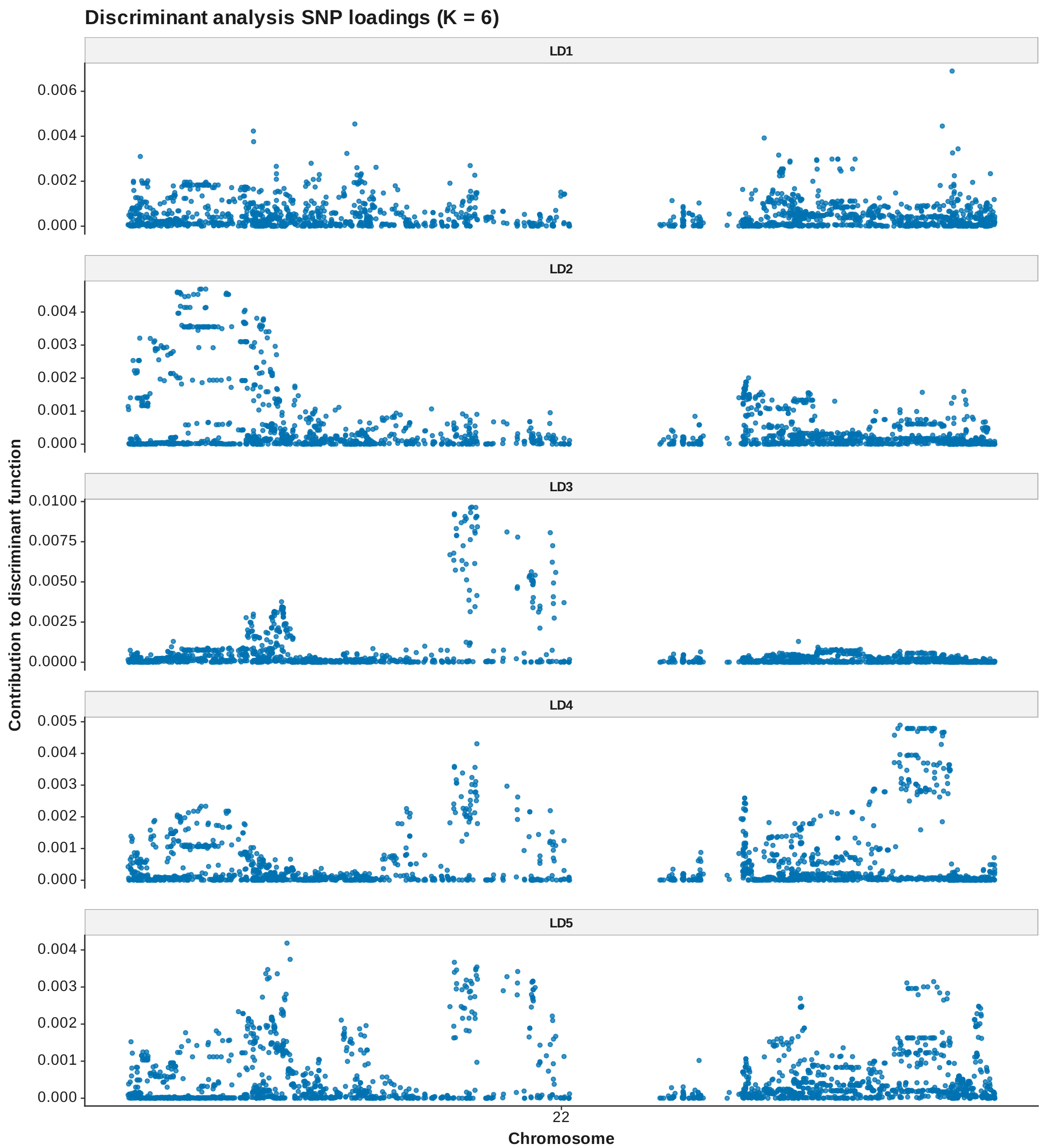

Figure 45: DAPC loadings manhattan K 6

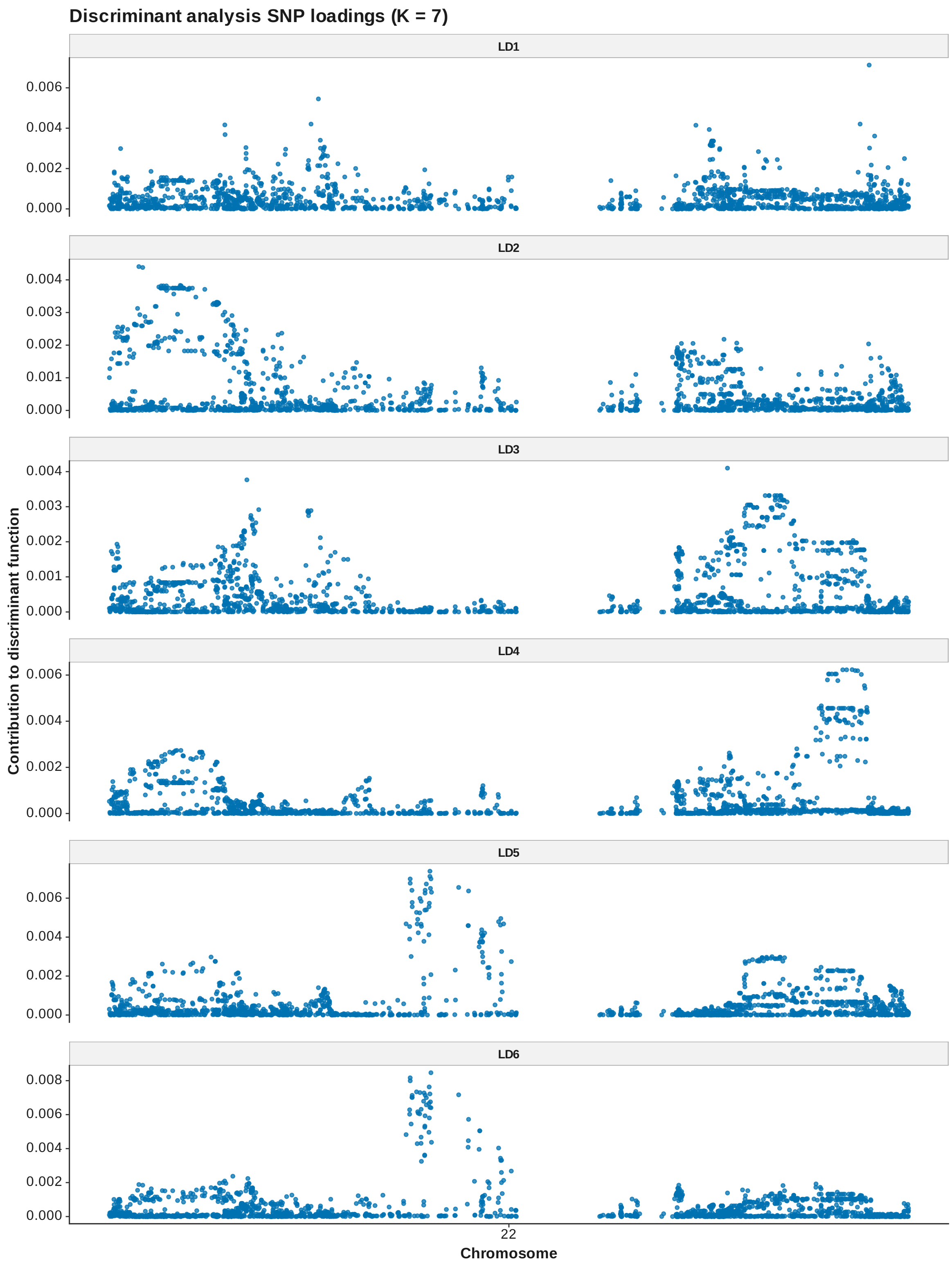

Figure 46: DAPC loadings manhattan K 7

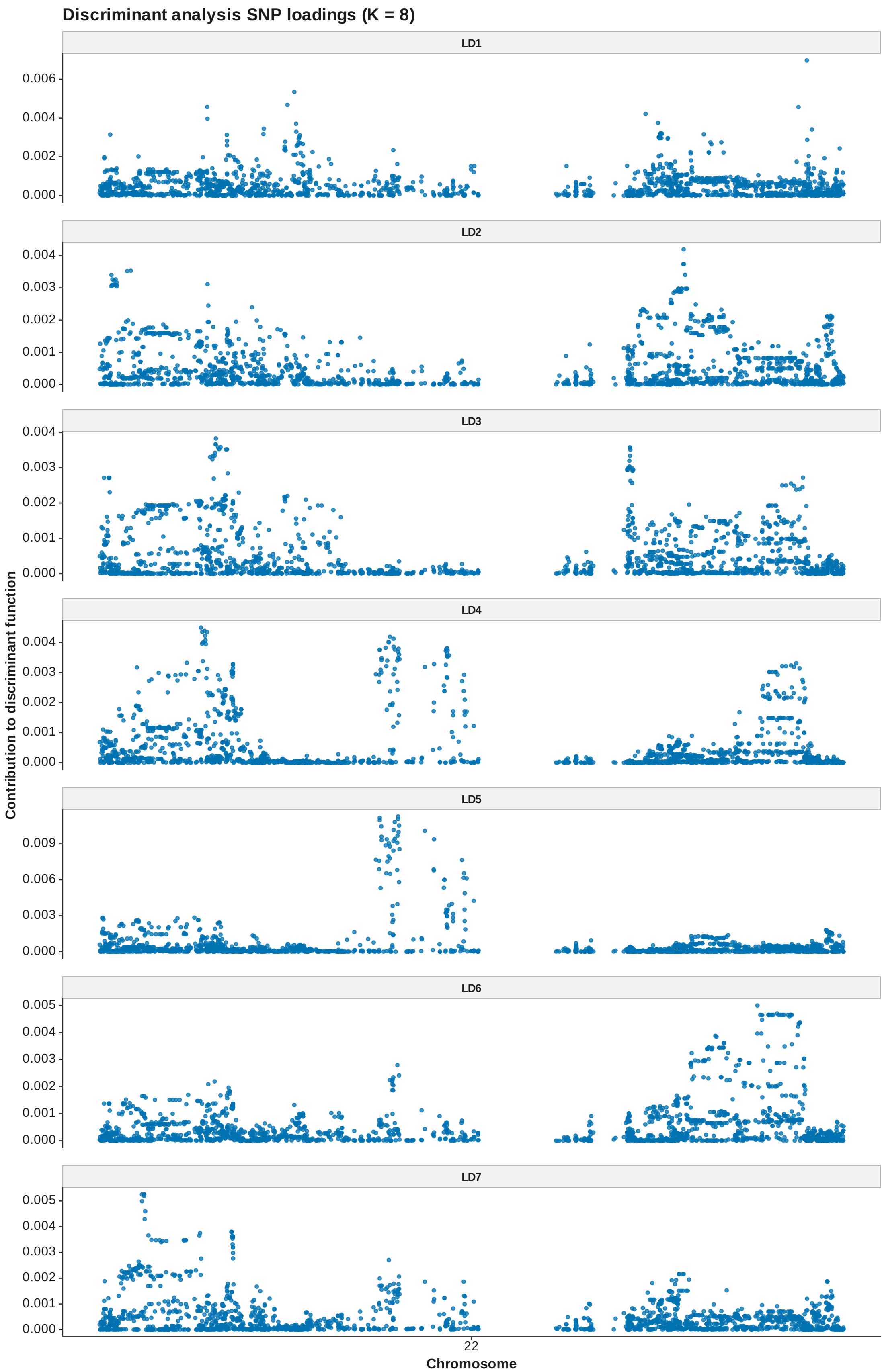

Figure 47: DAPC loadings manhattan K 8

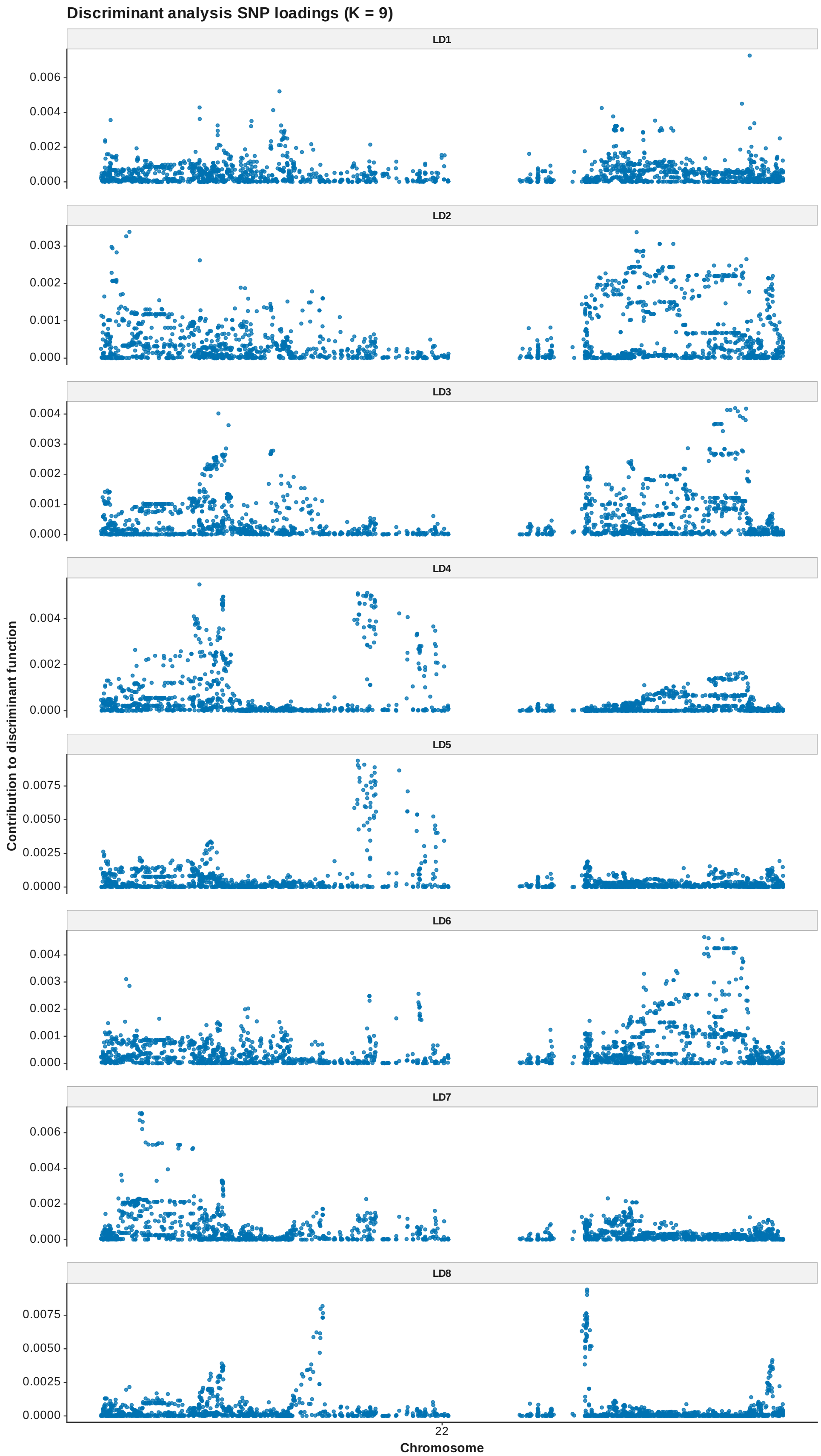

Figure 48: DAPC loadings manhattan K 9

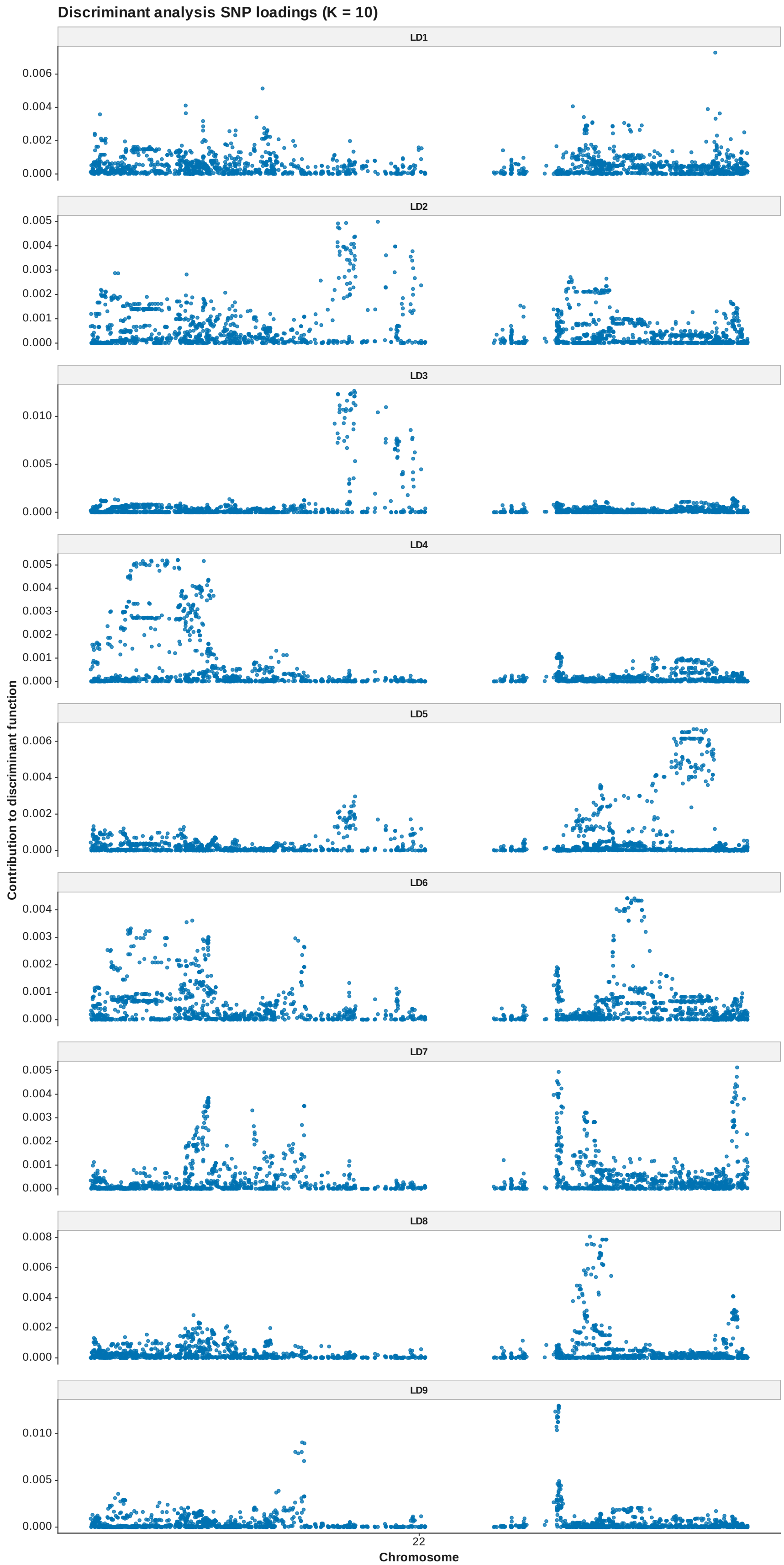

Figure 49: DAPC loadings manhattan K 10

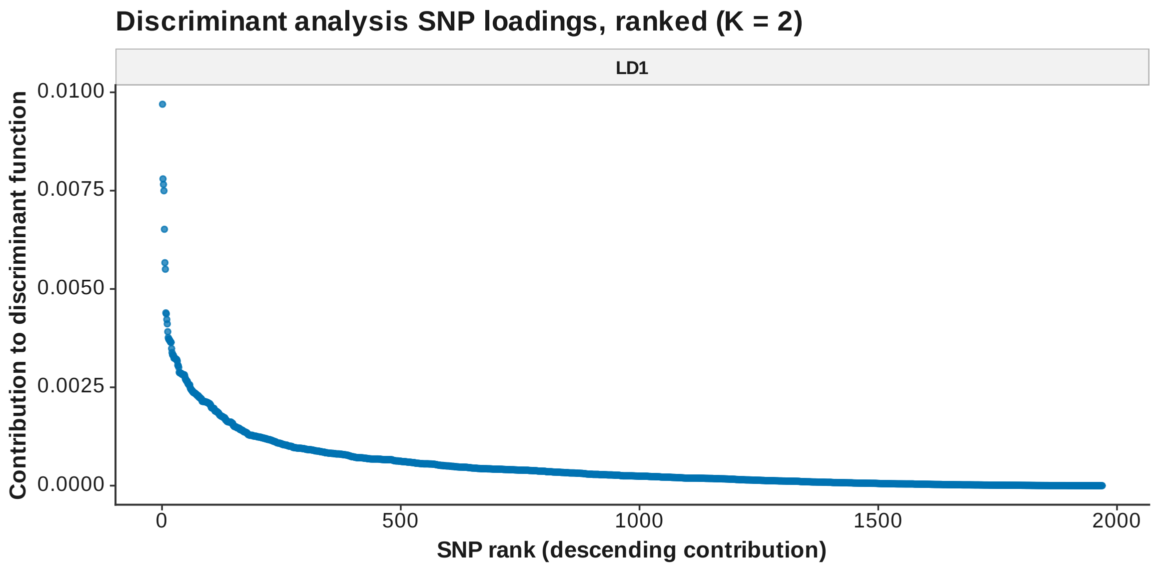

Figure 50: DAPC loadings ranked K 2

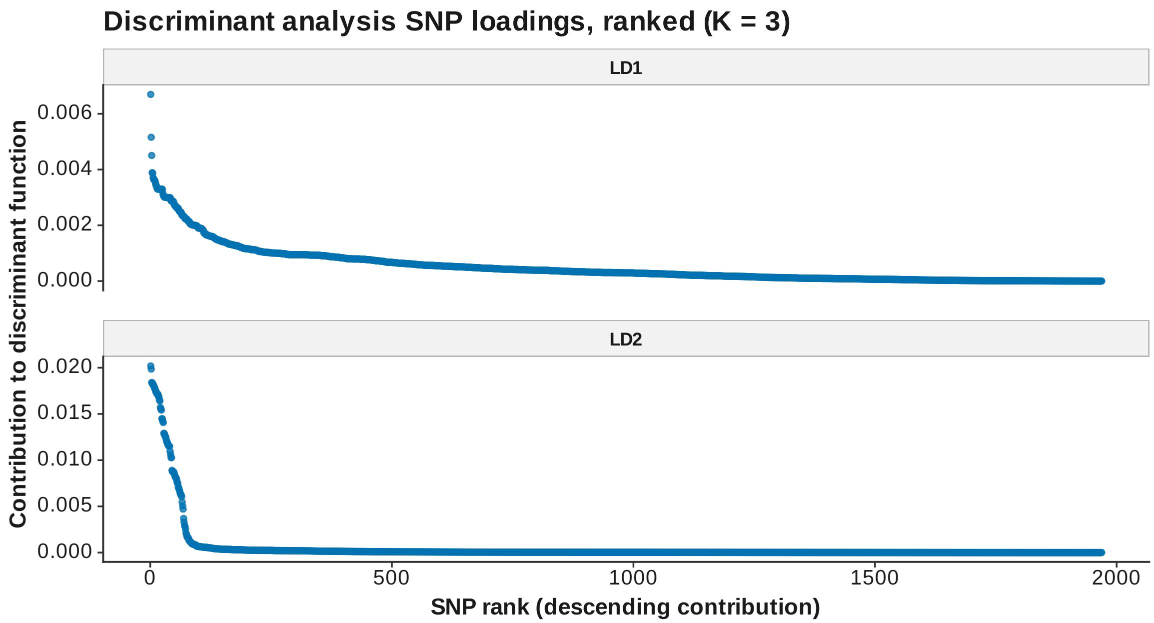

Figure 51: DAPC loadings ranked K 3

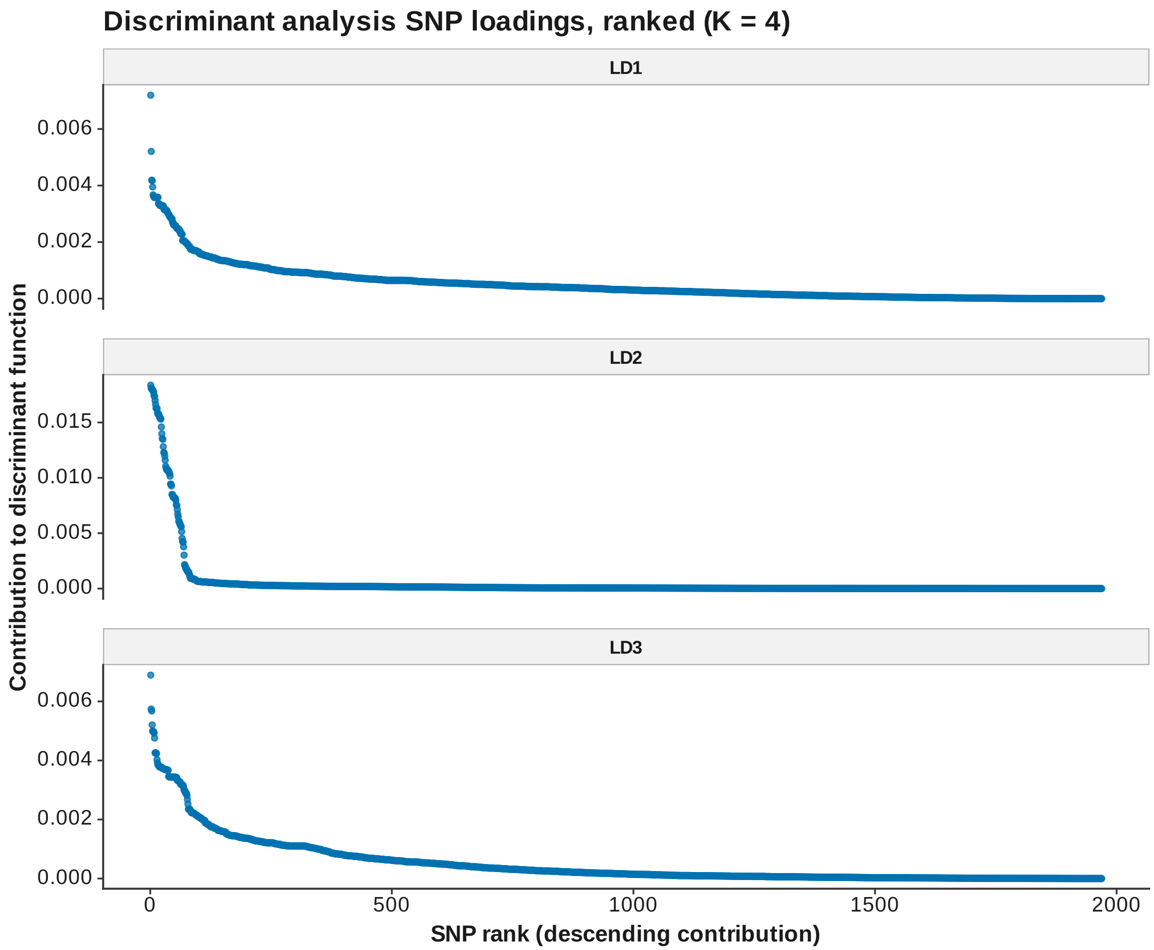

Figure 52: DAPC loadings ranked K 4

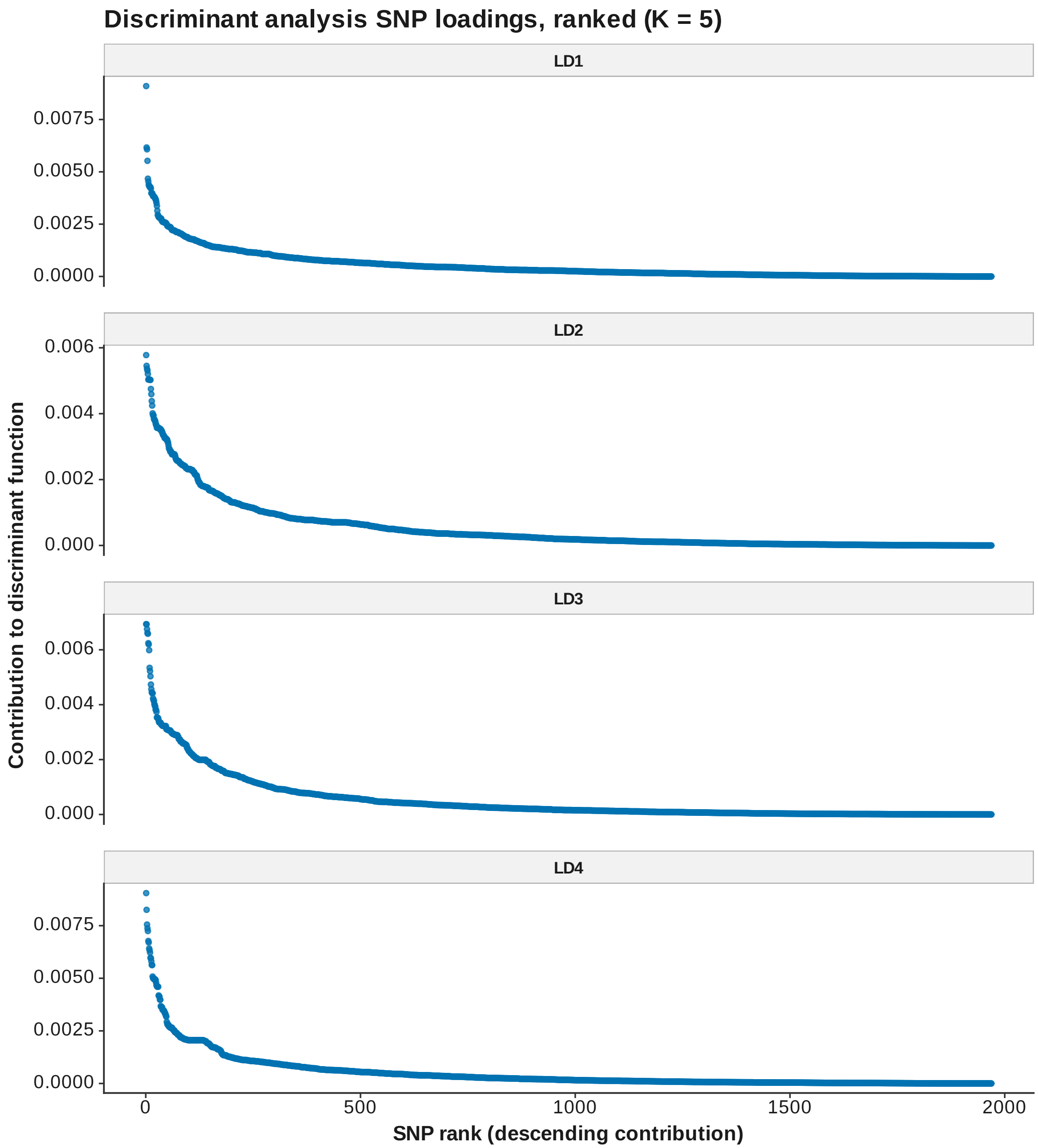

Figure 53: DAPC loadings ranked K 5

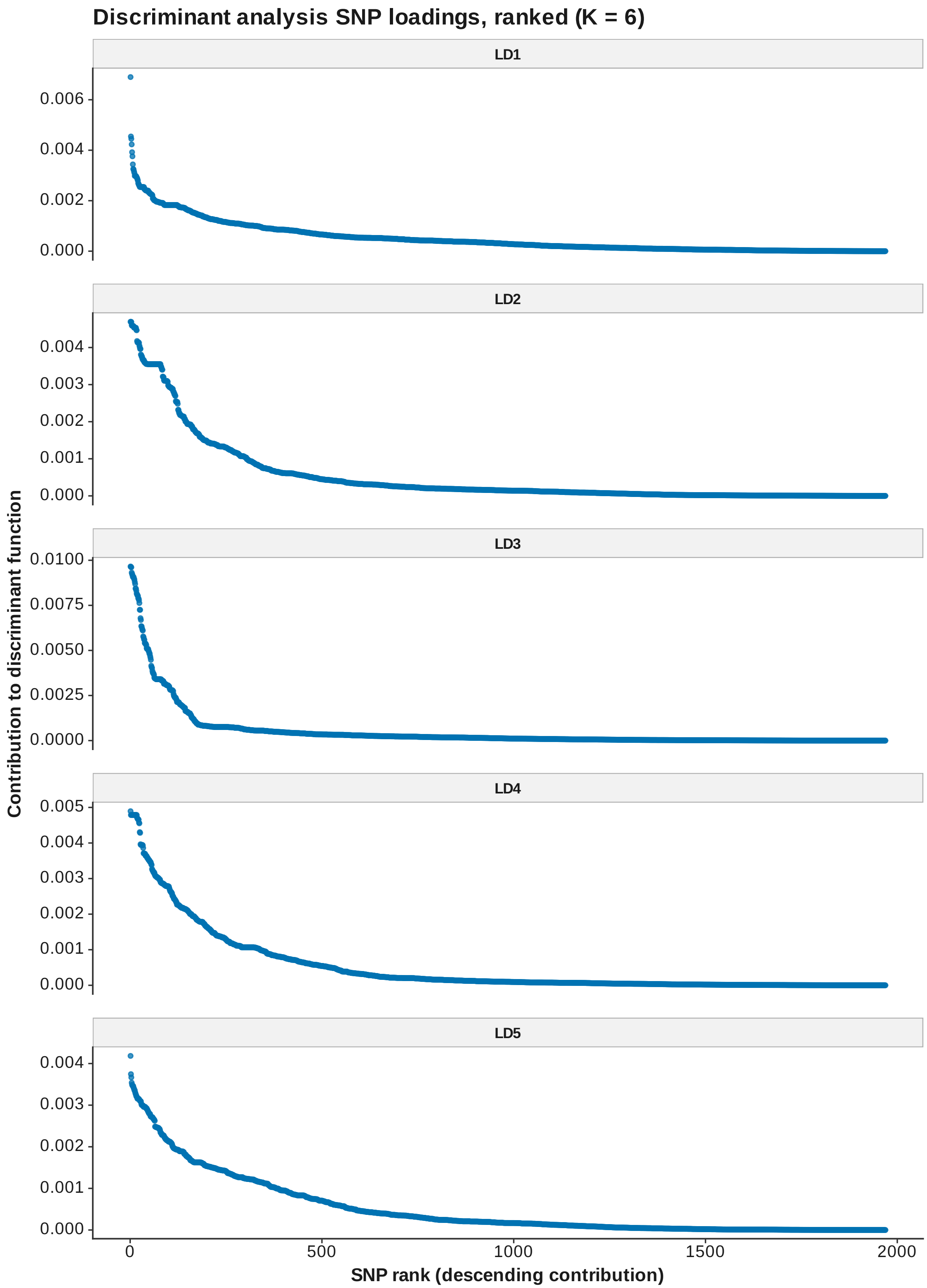

Figure 54: DAPC loadings ranked K 6

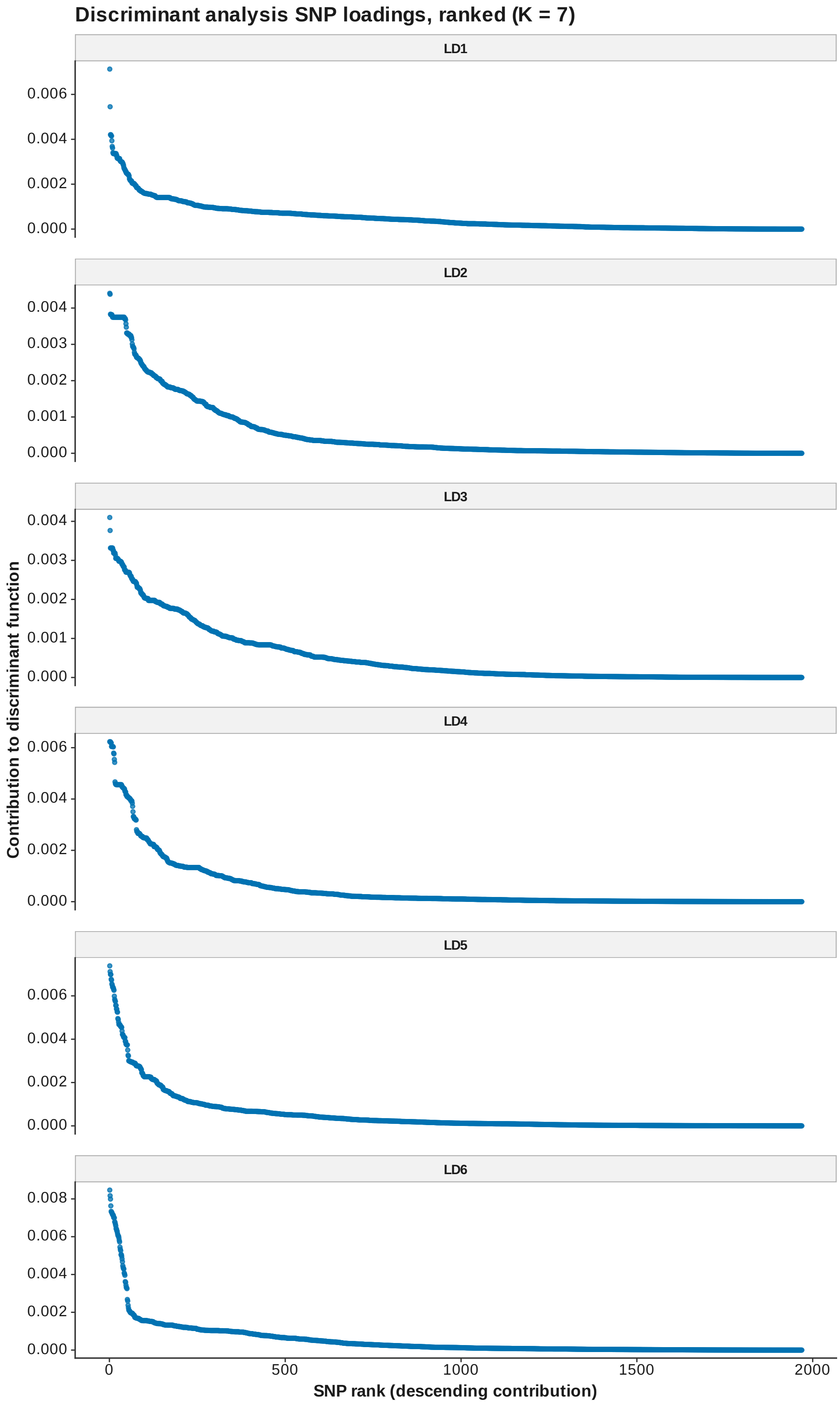

Figure 55: DAPC loadings ranked K 7

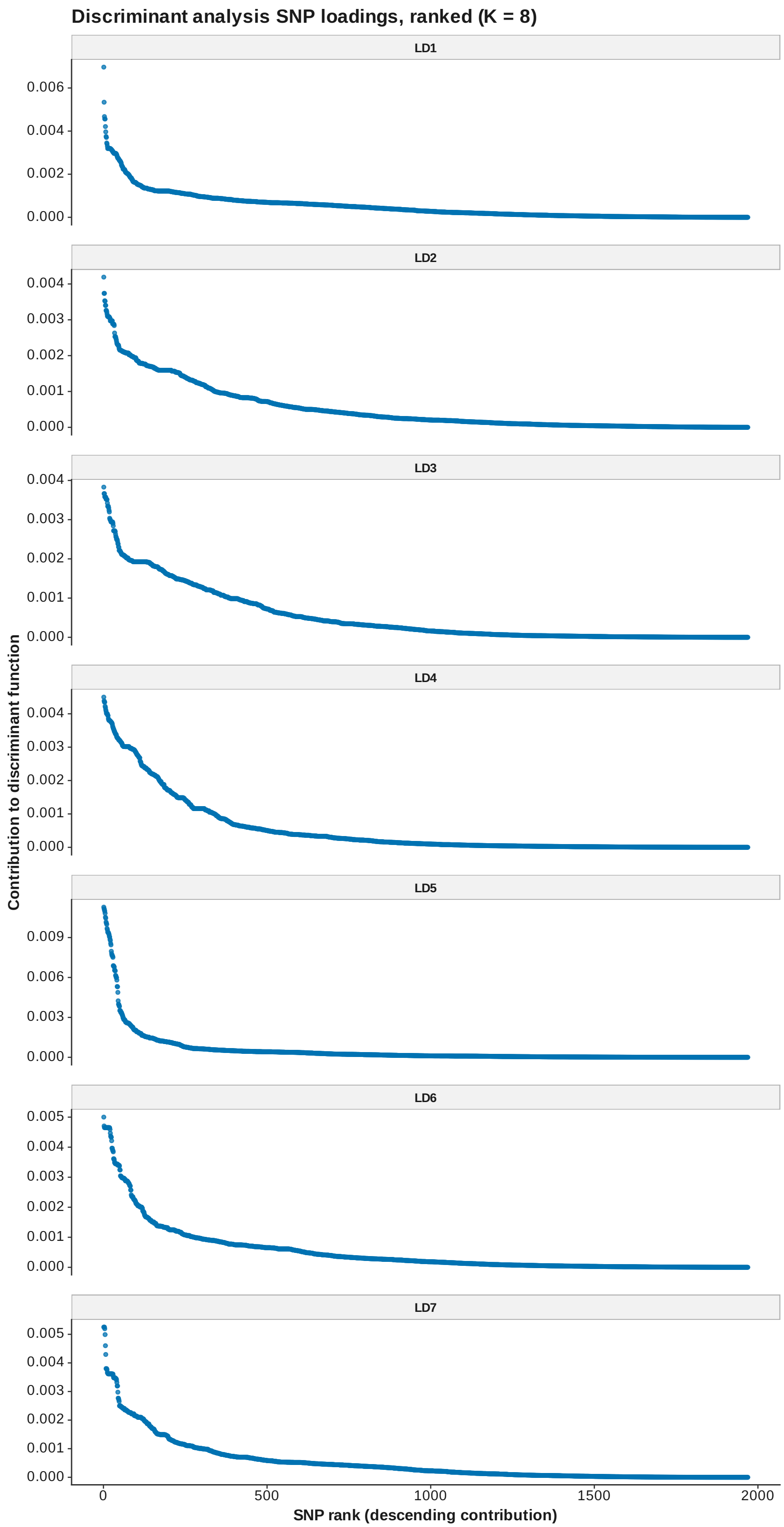

Figure 56: DAPC loadings ranked K 8

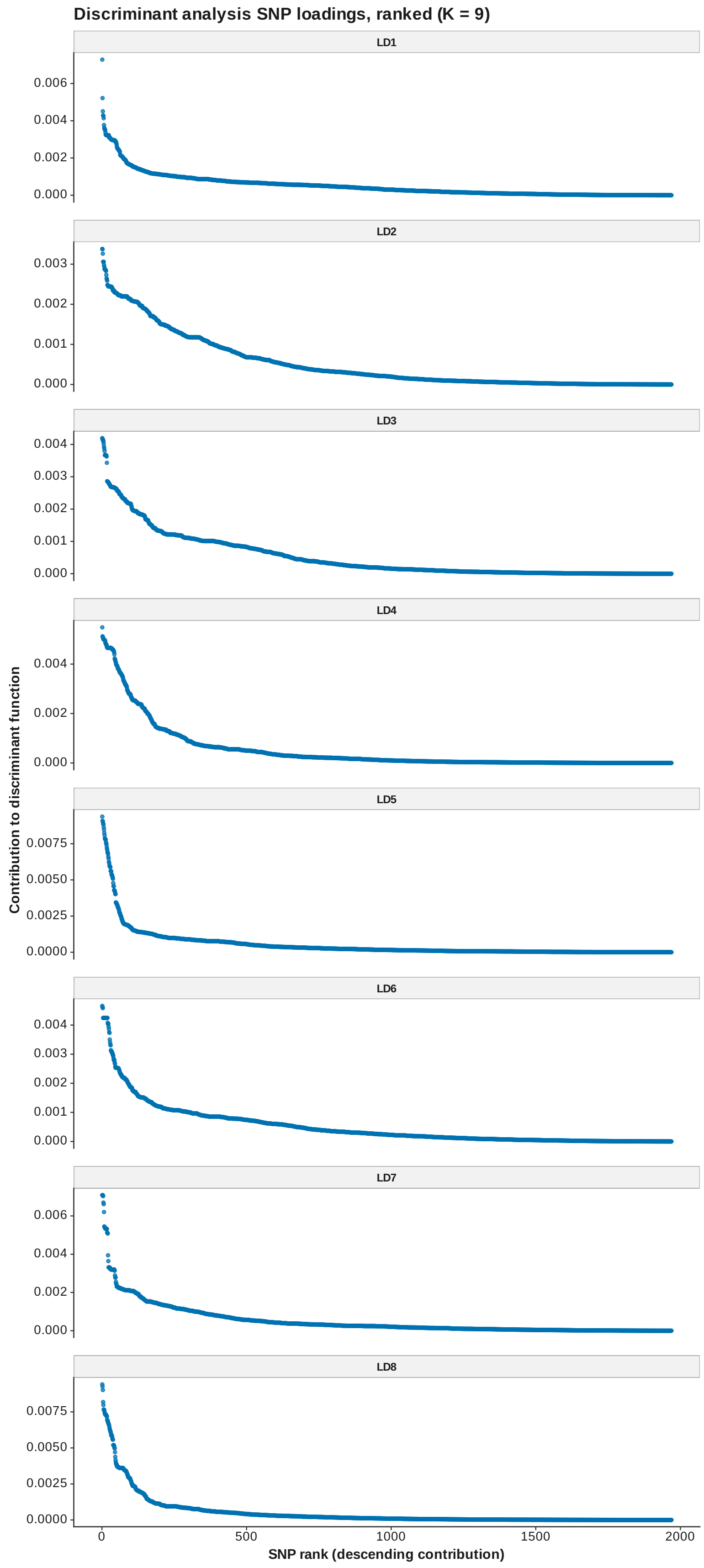

Figure 57: DAPC loadings ranked K 9

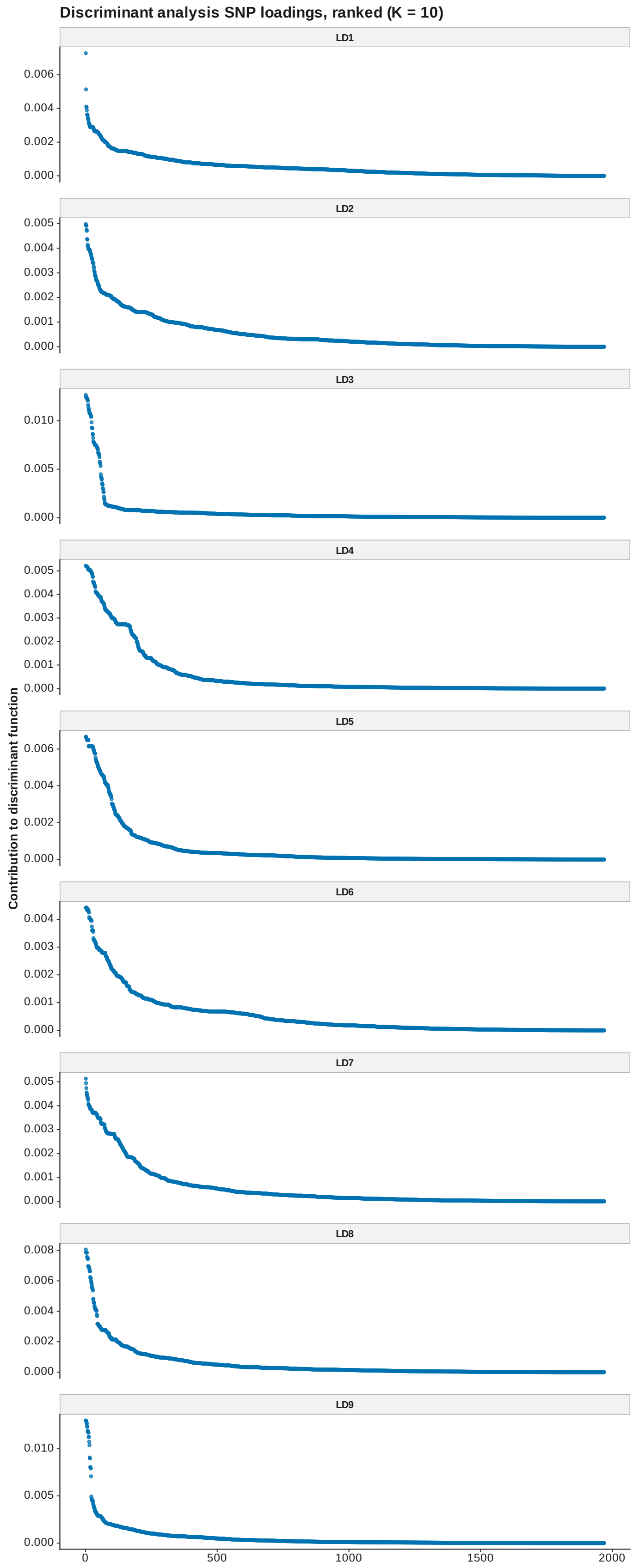

Figure 58: DAPC loadings ranked K 10

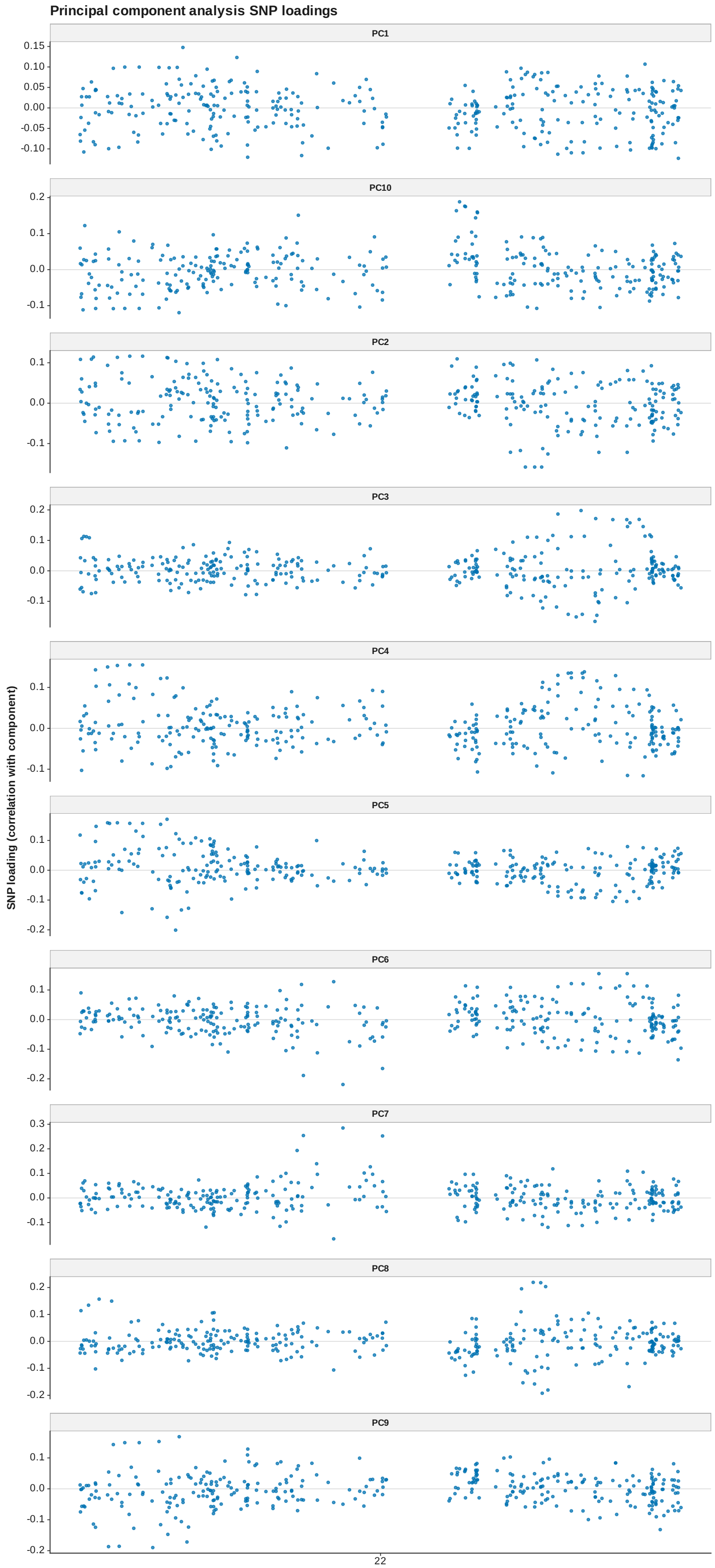

Figure 59: PCA loadings manhattan

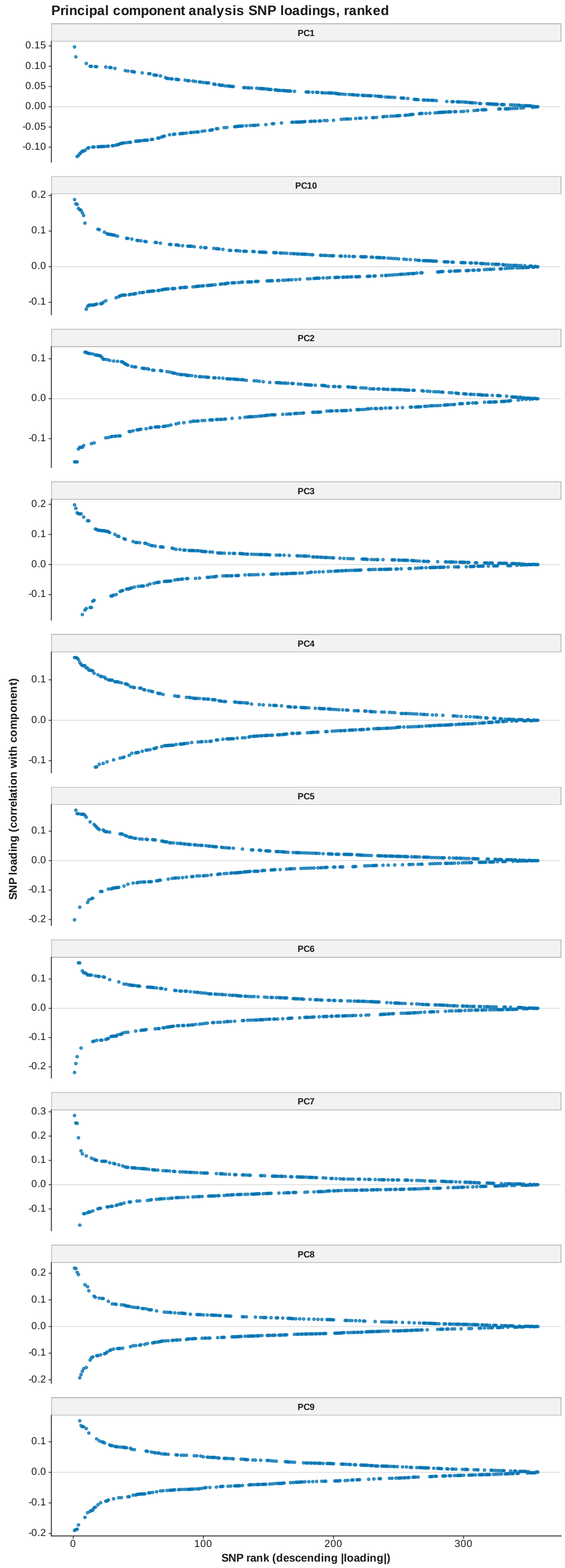

Figure 60: PCA loadings ranked

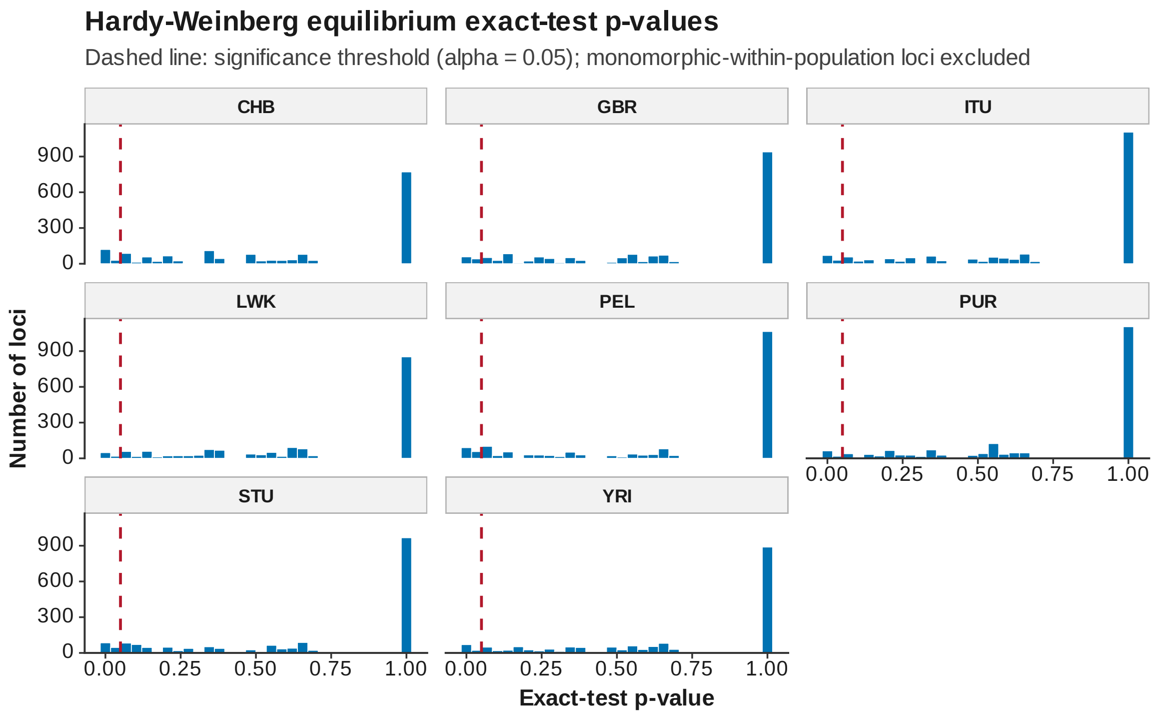

Figure 61: HWE pvalues

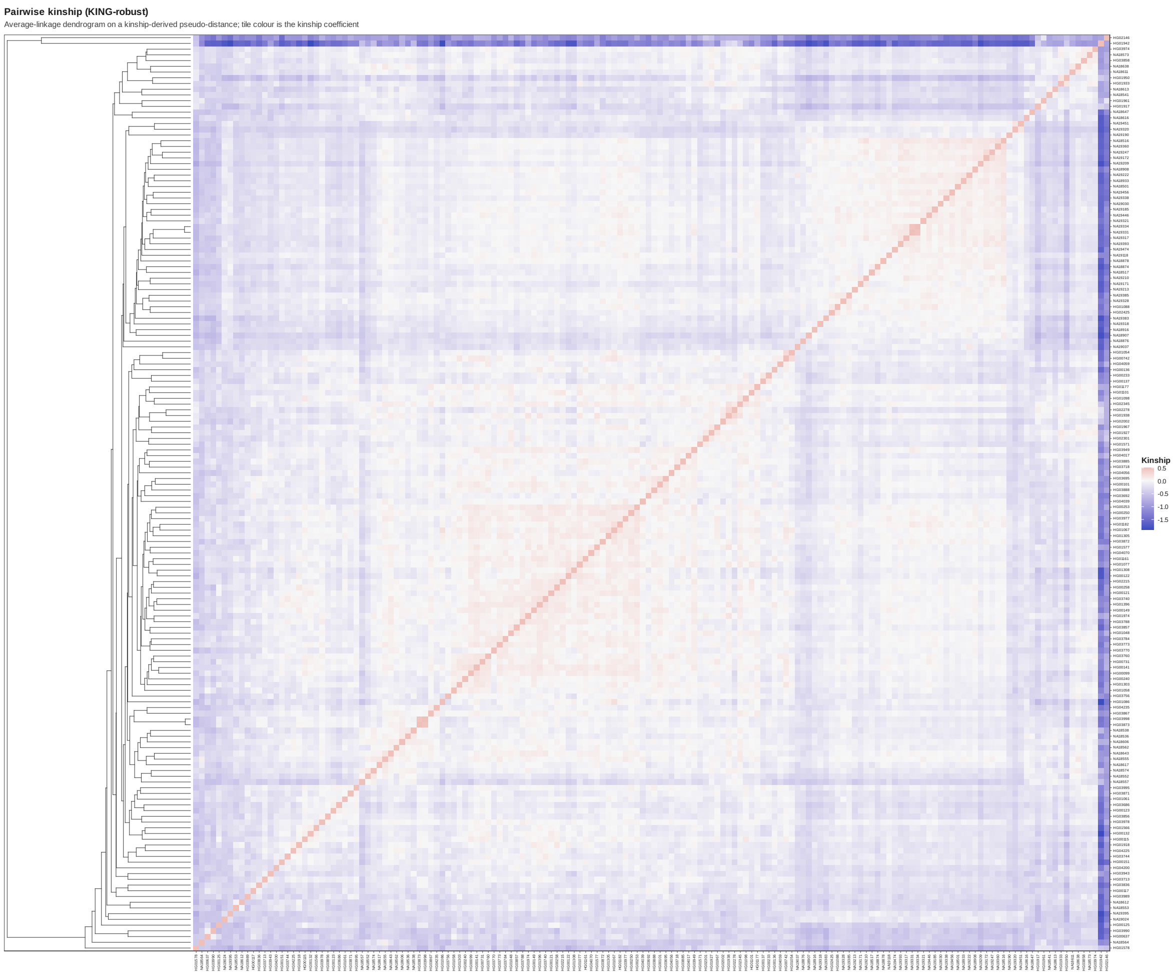

Figure 62: Kinship heatmap

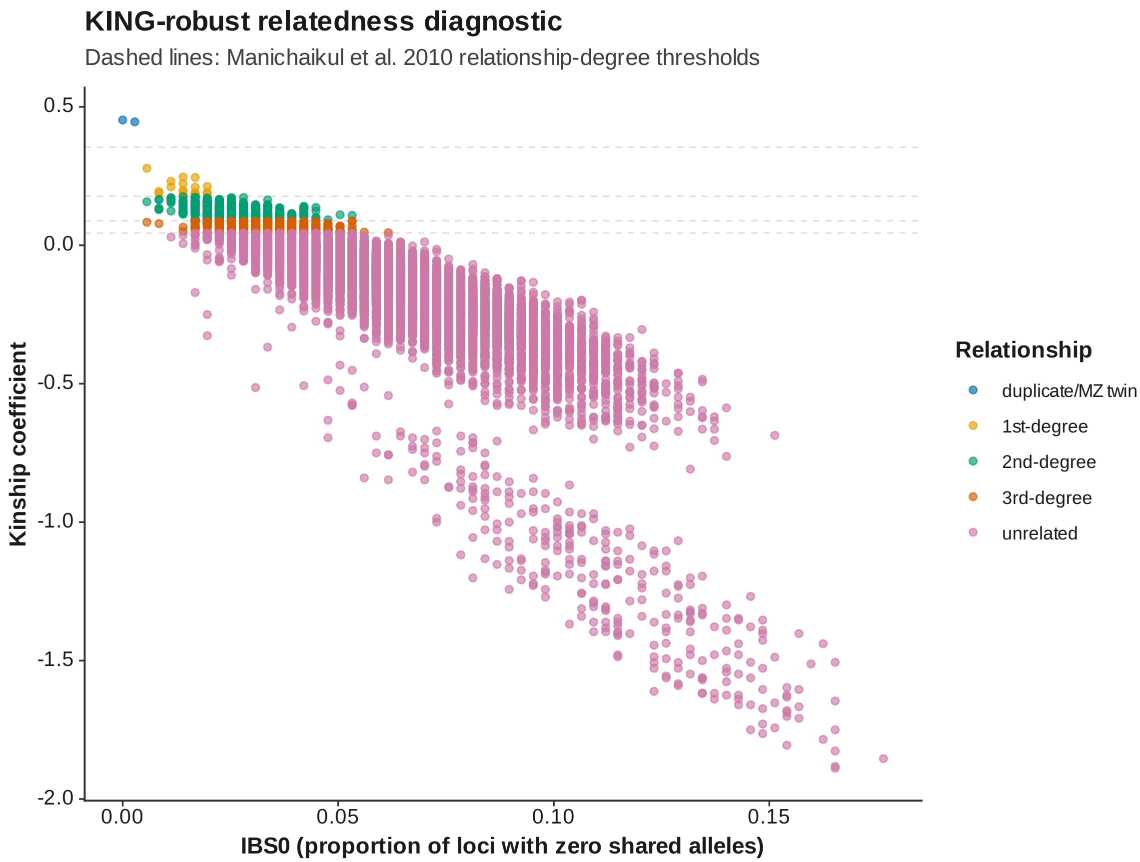

Figure 63: Kinship IBS 0 vs kinship

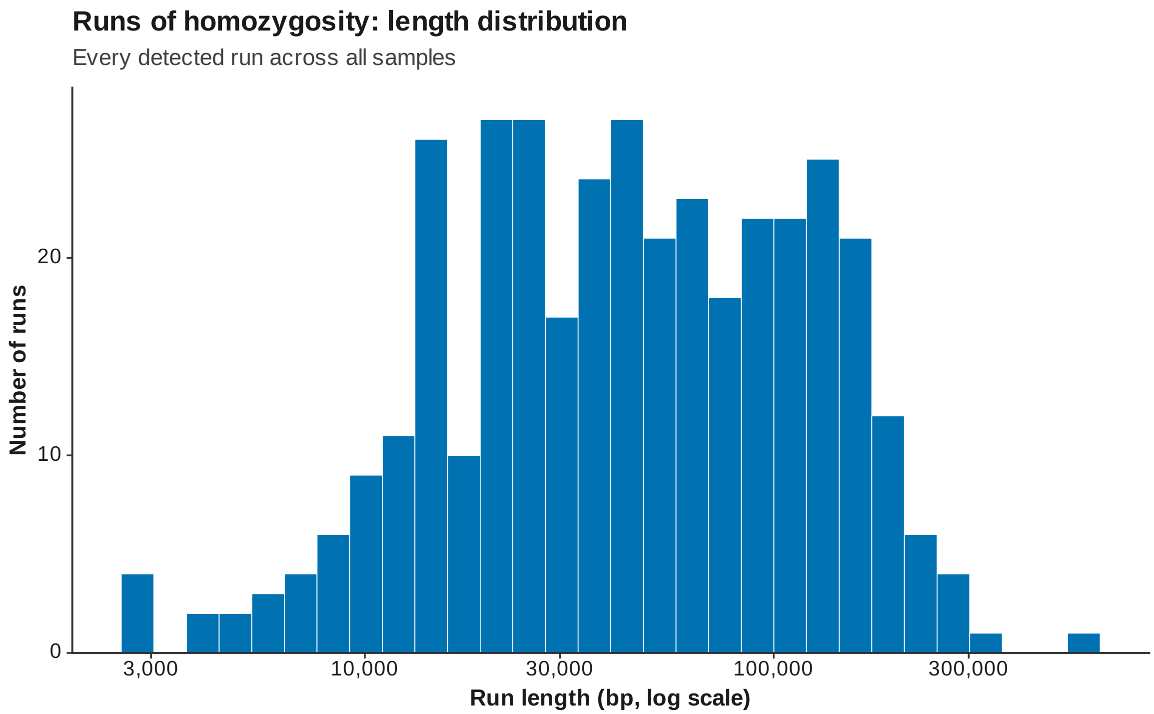

Figure 64: ROH length distribution

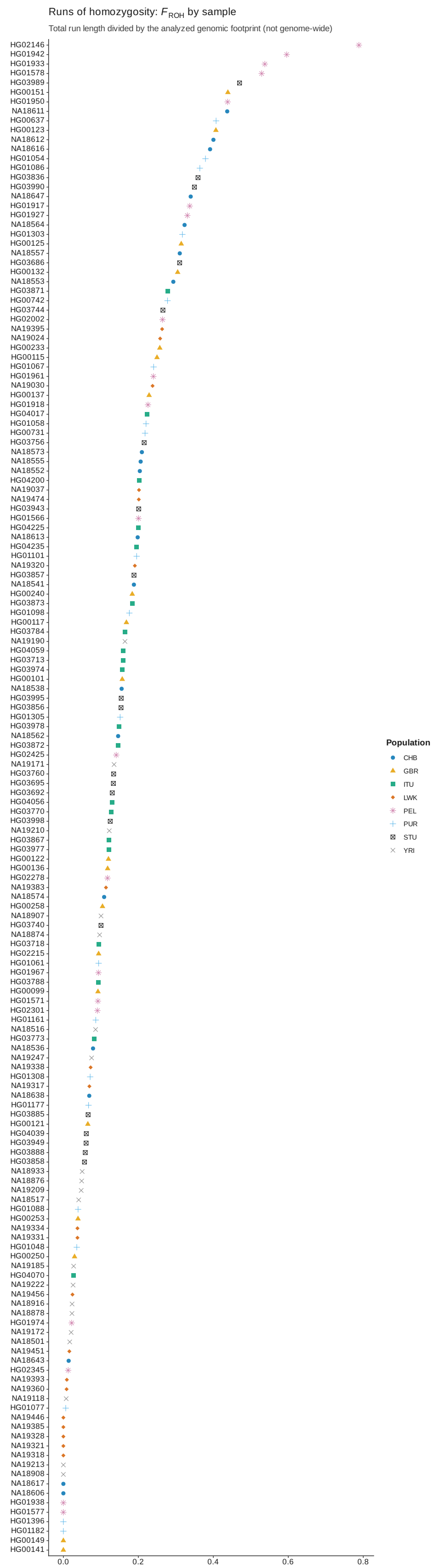

Figure 65: ROH FROH by sample

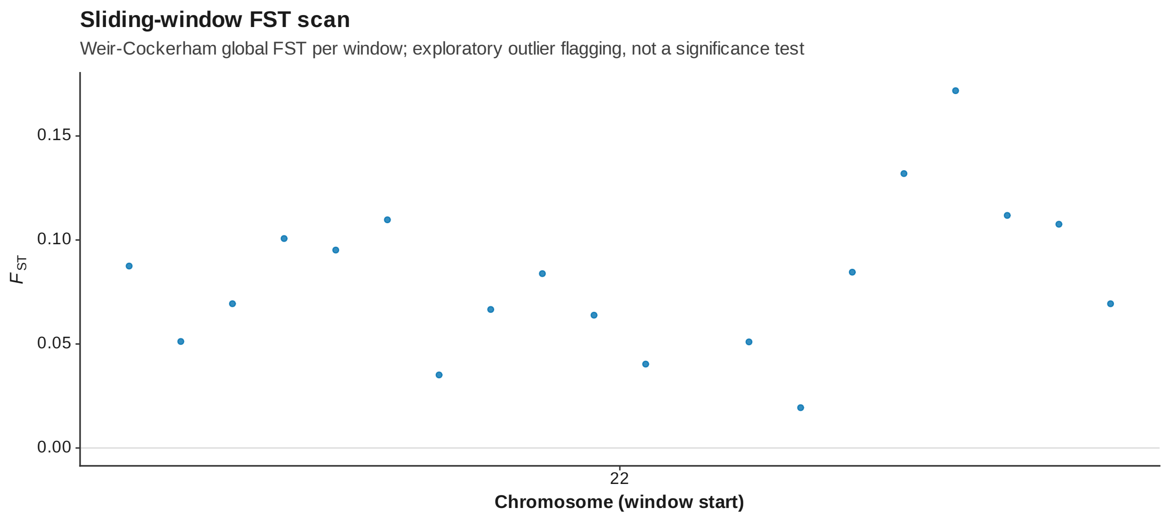

Figure 66: Genome scan FST manhattan

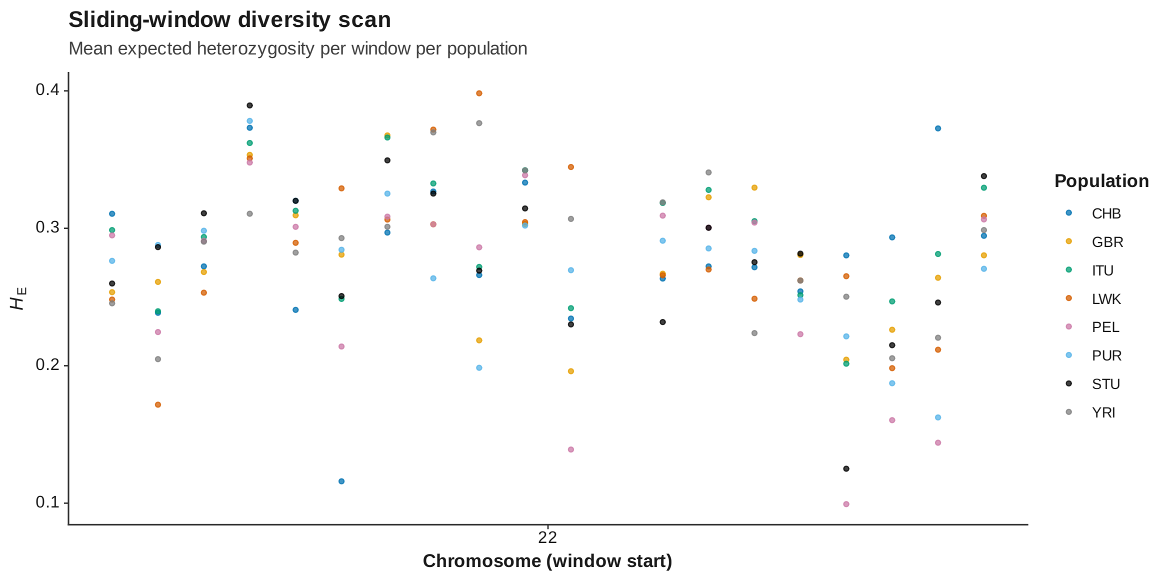

Figure 67: Genome scan diversity manhattan

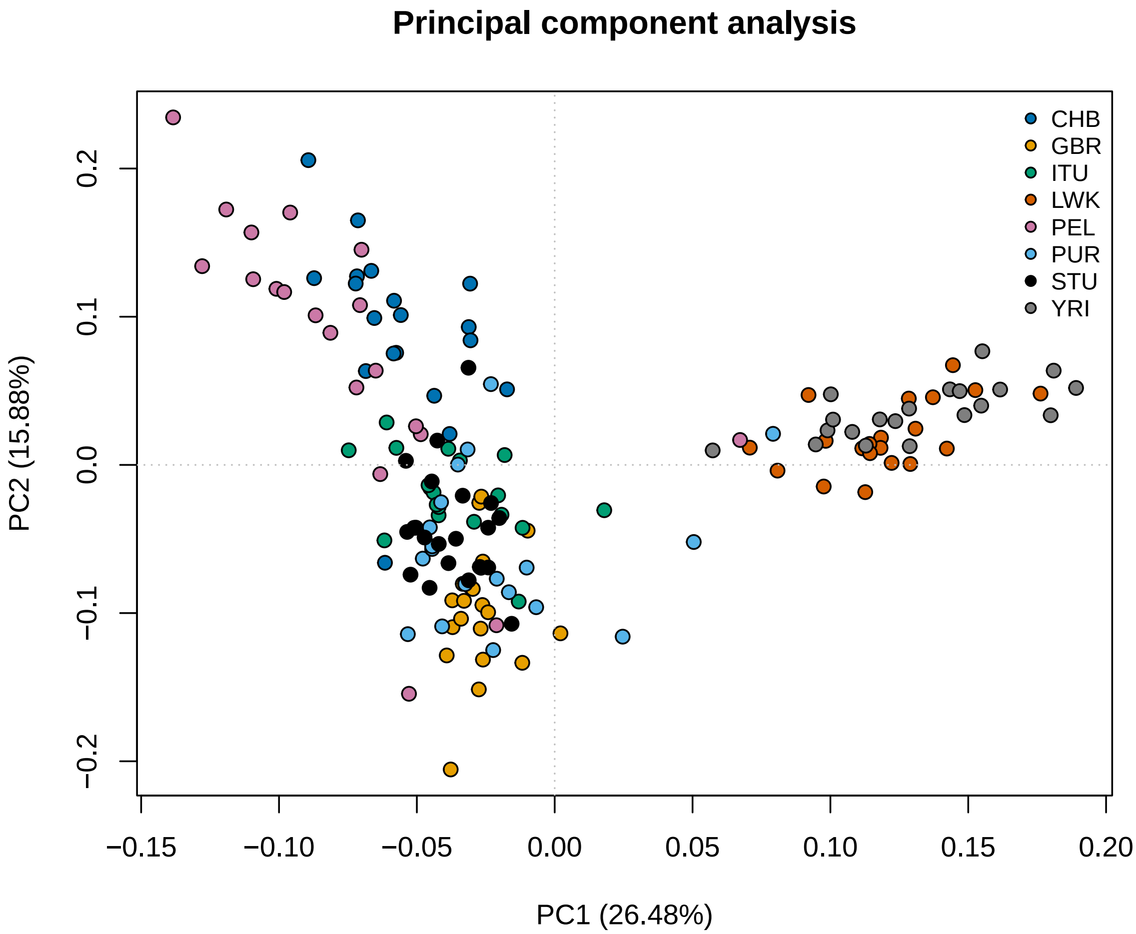

Figure 68: PCA PC 1 PC 2

16 Reproducibility

This report was generated from the serialized pipeline result object. Exact input hashes, settings, package versions, and stage runtimes are stored alongside the report.

16.1 Population-structure reproducibility

K	assignment_accuracy	replicate_max_rmse
2	0.2500	0.0971
3	0.2938	0.0000
4	0.3562	0.3841
5	0.4062	0.3512
6	0.3938	0.2944
7	0.4812	0.0727
8	0.4625	0.1614
9	0.4688	0.1830
10	0.3812	0.2373